EE345 – 第三代半导体基础导论
Introduction to Wide-bandgap Semiconductors

Chapter 06
GaN Transistors and applications

Review of last lecture

- Advantages and disadvantages of SiC power devices
- Design considerations for a power switch
- Current status of the 3rd generation semiconductor research
- Mobility and on-resistance of SiC MOSFET
- Fabrication process
- Packaging and failure analysis

Contents in this lecture

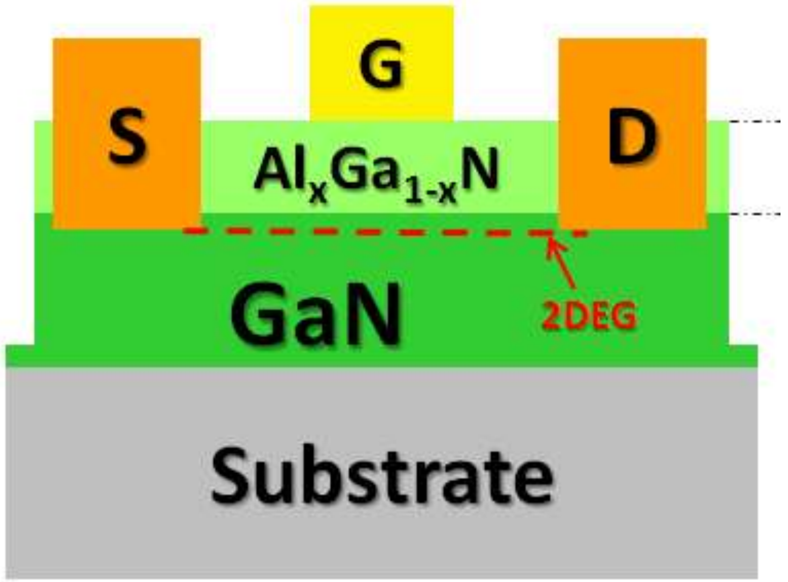
- GaN HEMTs (High Electron Mobility Transistors)

Historical development of GaN HEMTs

Year	Event	Authors
1969	GaN by hydride vapor phase epitaxy	Maruska and Tietjen
1971	GaN by MOCVD	Manasevit et al.
1992	AlGaN/GaN two-dimensional electron gas	Khan et al.
1993	AlGaN/GaN HEMT	Khan et al.
1994	Microwave AlGaN/GaN HFET	Khan et al.
1996	Microwave power AlGaN/GaN MODFET	Wu et al.
1999	Reveal current compression in GaN MODFET	Kohn et al.
1999	6.9 W/mm @ 10 GHz GaN HEMT on SiC	Sheppard et al.
2000	Surface passivated AlGaN/GaN HEMTs	Green et al.
2004	30 W/mm @ 8 GHz GaN HEMT with field plate	Wu et al.
2005	High performance enhancement-mode HEMT	Cai et al.

GaN HEMTs

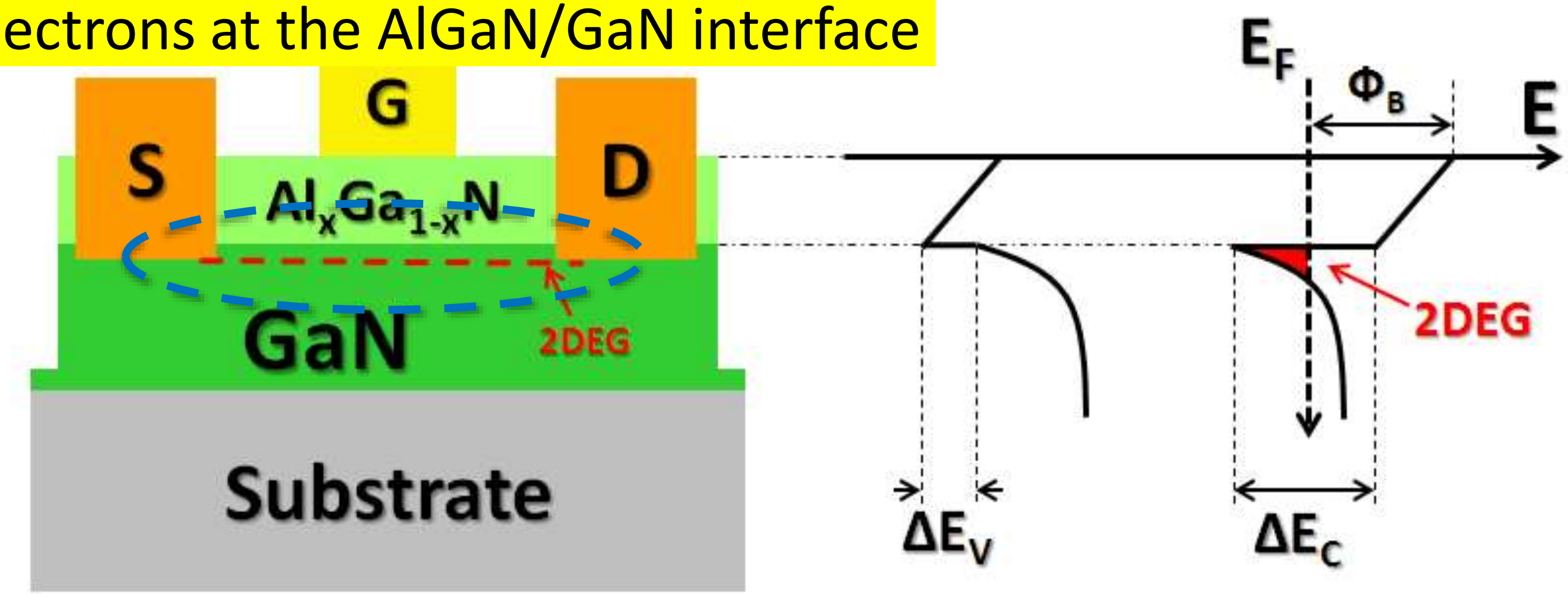
– 4 terminals device
– Normally-on (depletion mode)
– Heterostructure junction based
– High electron density in the channel (2DEG)
– Gate-to-drain distance withstand the high electrical field in the off-state



The cross-sectional schematic and the energy-band profile of a typical $Al_xGa_{1-x}N$/GaN HEMT.

GaN HEMTs

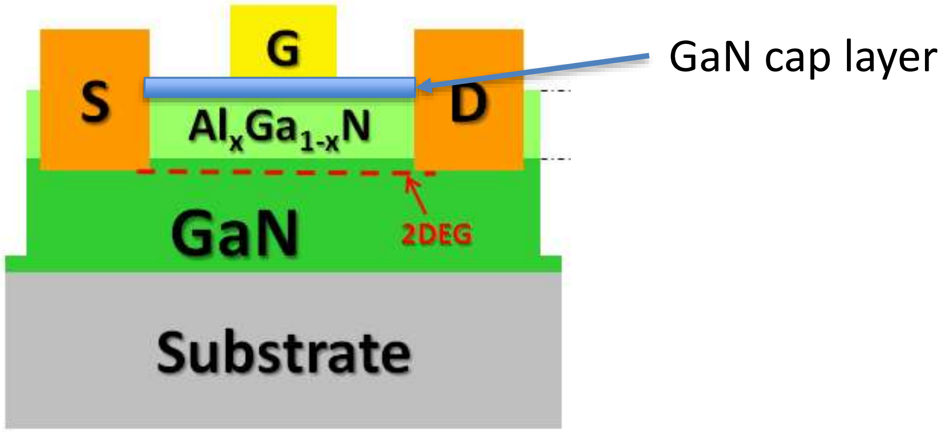
- High electron mobility transistors (HEMTs)
  - $Al_xGa_{1-x}N$/GaN heterostructure
  - The core of the HEMT
  - Thickness and doping of the GaN layer affect the off-state performance
  - 2DEG: 2-dimensional electron gas (Why gas?)
  - 2DEG： confinement of electrons at the AlGaN/GaN interface



The cross-sectional schematic and the energy-band profile of a typical $Al_xGa_{1-x}N$/GaN HEMT.

GaN HEMTs

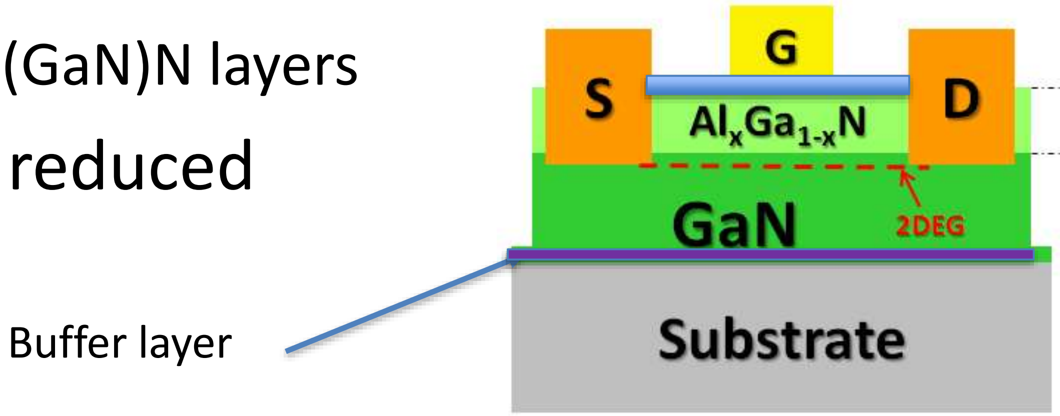
- GaN-cap layer
  - Reduce the reverse leakage current through the Schottky contact
  - Increase in the electric field strength in the AlGaN layer in the off-state so improve reliability
  - Increase power efficiency by increasing the thickness



The cross-sectional schematic and the energy-band profile of a typical $Al_xGa_{1-x}N$/GaN HEMT.

GaN HEMTs

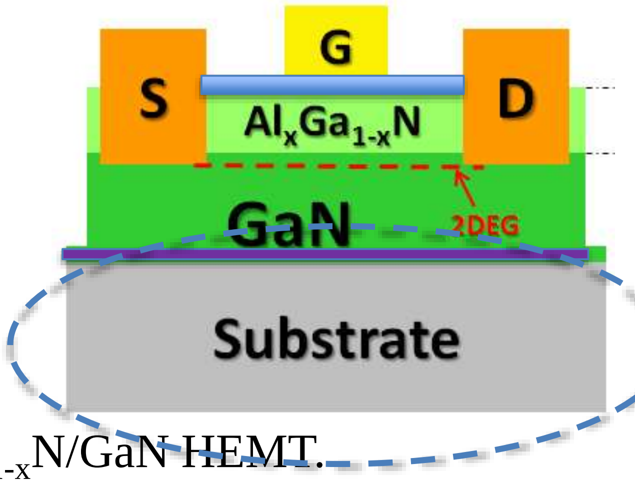
- Buffer
  - Function: it affects the structural and optical properties of the GaN layer and the on- and off-state electrical characterizations of the device
  - Most common used buffer layers:
    - Single layer of AlN and AlGaN
    - Superlattice structure made of GaN and Al(GaN)N layers
  - Increasing the buffer thickness yields a reduced substrate leakage



The cross-sectional schematic and the energy-band profile of a typical $Al_xGa_{1-x}N$/GaN HEMT.

GaN HEMTs - Substrates

- Possible substrate for GaN HEMTs:
- Sapphire (Al_2O_3)
  - Pros: Semi-insulating, can withstand high growth temperature, relatively cheap
  - Cons: Very low thermal conduction (how low?) , large lattice mismatch (how large?)
- Silicon carbide (SiC)
  - Pros: High thermal conductivity, low lattice mismatch
  - Cons Cost, crystallographic defects
- Silicon (Si)
  - Pros: Low cost, excellent availability of large diameters, acceptable thermal conductivity, processing in standard silicon fabs (might not be the truth, why?)
  - Cons: Large lattice mismatch

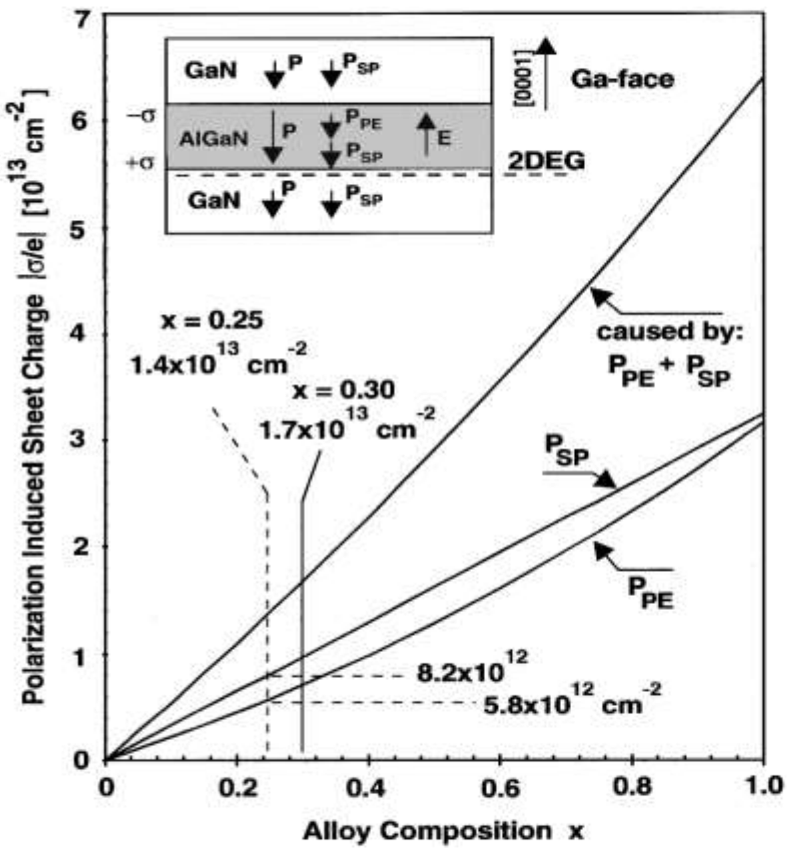


The cross-sectional schematic and the energy-band profile of a typical $Al_xGa_{1-x}N$/GaN HEMT.

GaN HEMTs fundamentals

- Polarization Effect
  - spontaneous polarization
  - piezoelectric polarization

- Two-dimensional electron gas (2DEG)
  - bulk scattering effects such as ionized impurity scattering are reduced (Recall the mobility model) ➔ high electron mobility



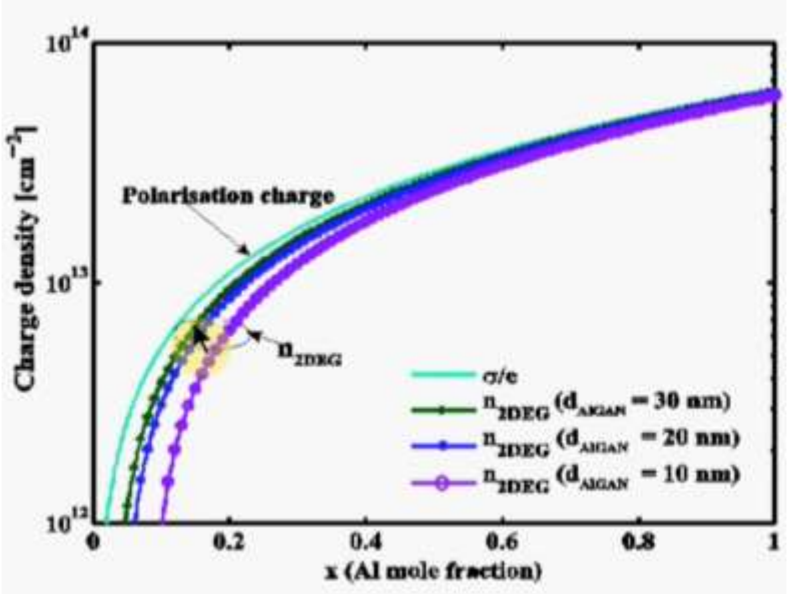
The calculated charge density at the $Al_xGa_{1-x}N$/GaN interface

2DEG dependence on thickness and x

- The 2DEG concentration comes from the solution of the Poisson equation (Gauss' law) across the energy band diagram of the $Al_xGa_{1-x}N/GaN$ heterojunction

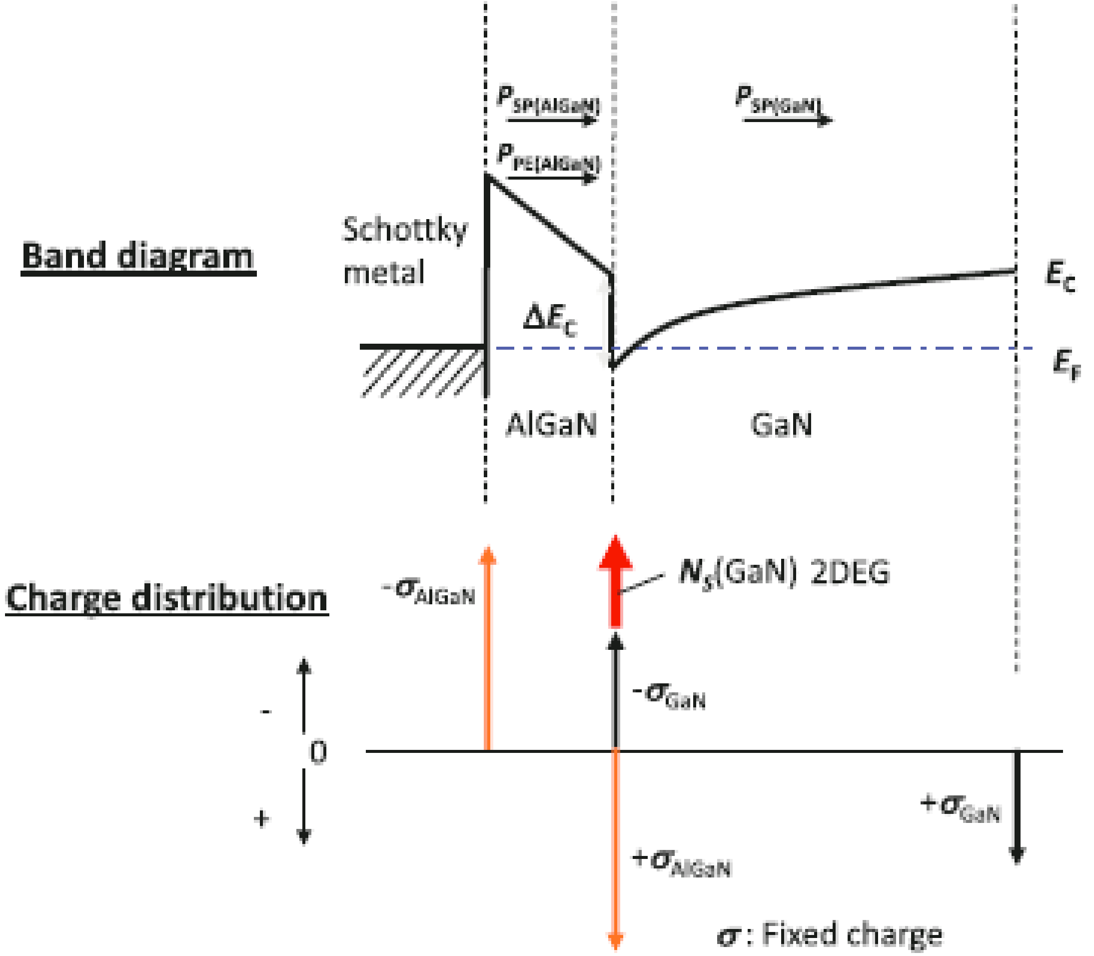
$$n_{2DEG} = \frac{\sigma}{e} - \left(\frac{\varepsilon_0 \varepsilon_r(x)}{d_{AlGaN} e^2}\right)[e\varphi_b(x) + E_{CF}(x) - \Delta E_c(x)]$$

- By increasing the thickness of the AlGaN layer the 2DEG density is enhanced



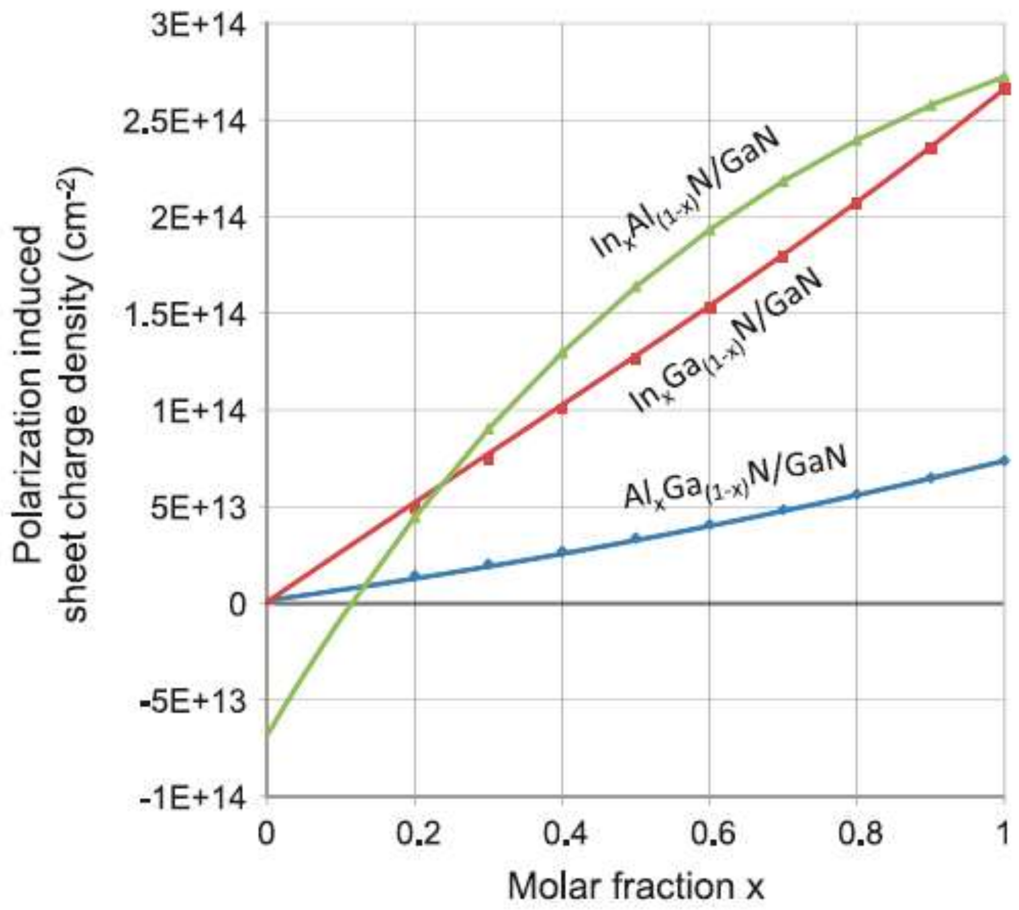
Band diagram

$$|\sigma(x)| = |P_{PE}(Al_X Ga_{1-x}\mathrm{N}) + P_{SP}(Al_X Ga_{1-x}\mathrm{N}) - P_{SP}(\mathrm{GaN})|$$



Band diagram of Schottky junction formed over $Al_{0.25}Ga_{0.75}N/GaN$

Polarization induced Carrier density

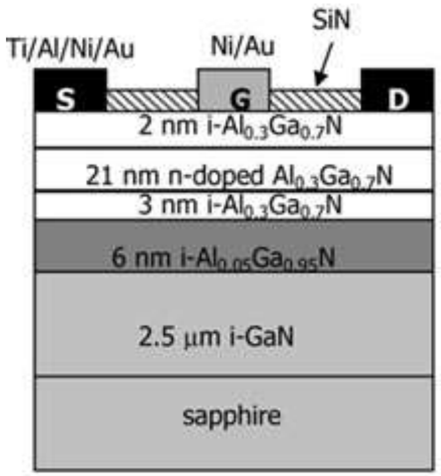


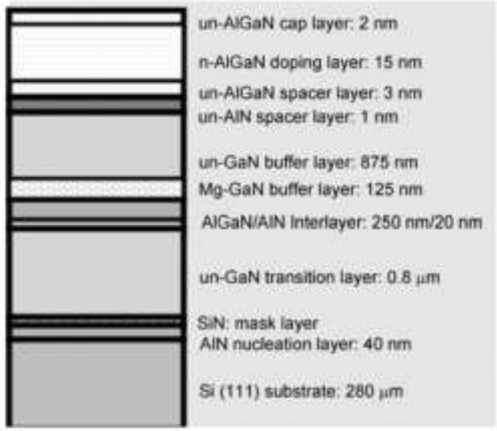
Theoretically calculated carrier densities at the heterostructure by spontaneous and piezoelectric polarizations

InAlN/GaN HEMTs: $In_{0.16}Al_{0.84}N$ can be lattice matched to GaN keeping high 2DEG density without piezoelectric effect, which is being studied at present.

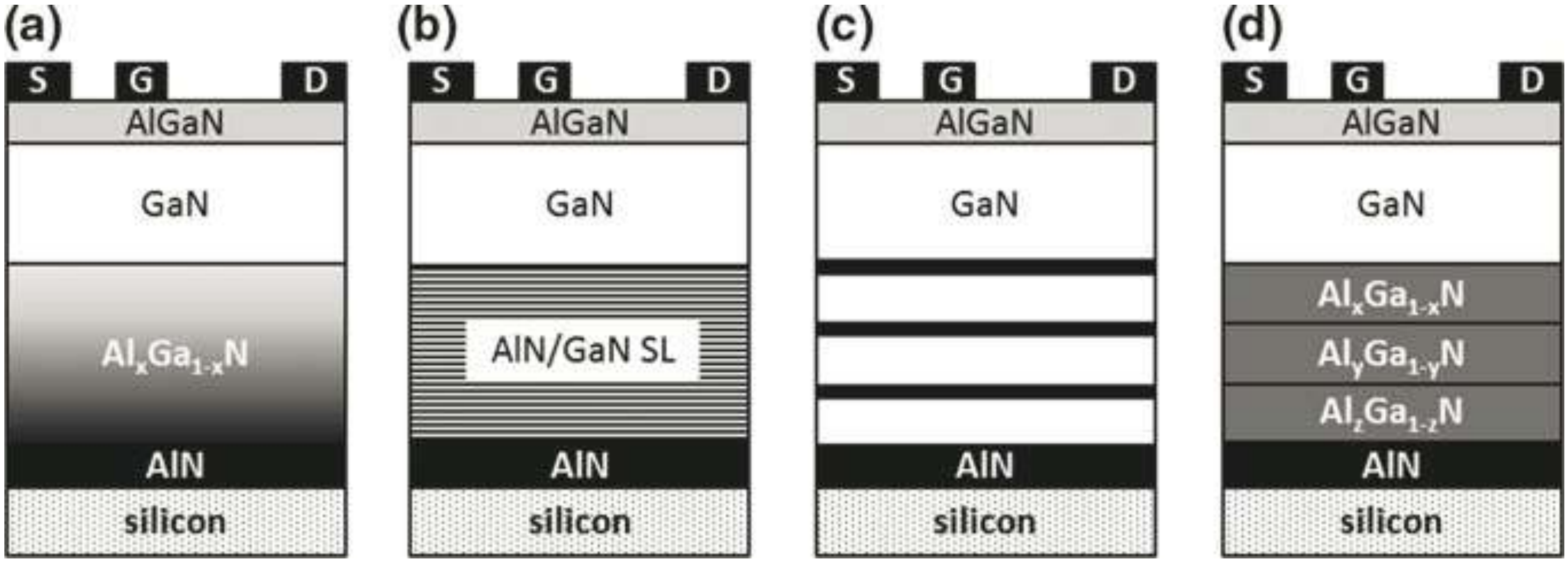
Epitaxy of AlGaN/GaN heterostructure - I

- MOCVD or MBE
  - Substrate (Si, SiC, Sapphire) → nucleation layer → buffer (isolation) layer → channel layer → barrier layer
  - Issues: crystalline quality, layer thickness, strain management, background concentration, interface/surface control …





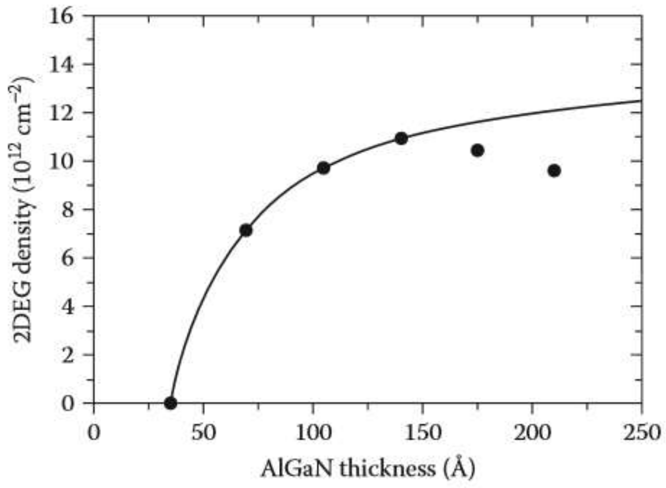
Epitaxy of AlGaN/GaN heterostructure - II



Schematics of transistors with different base layers for strain management on Si substrates:

Epitaxy of AlGaN/GaN heterostructure - III

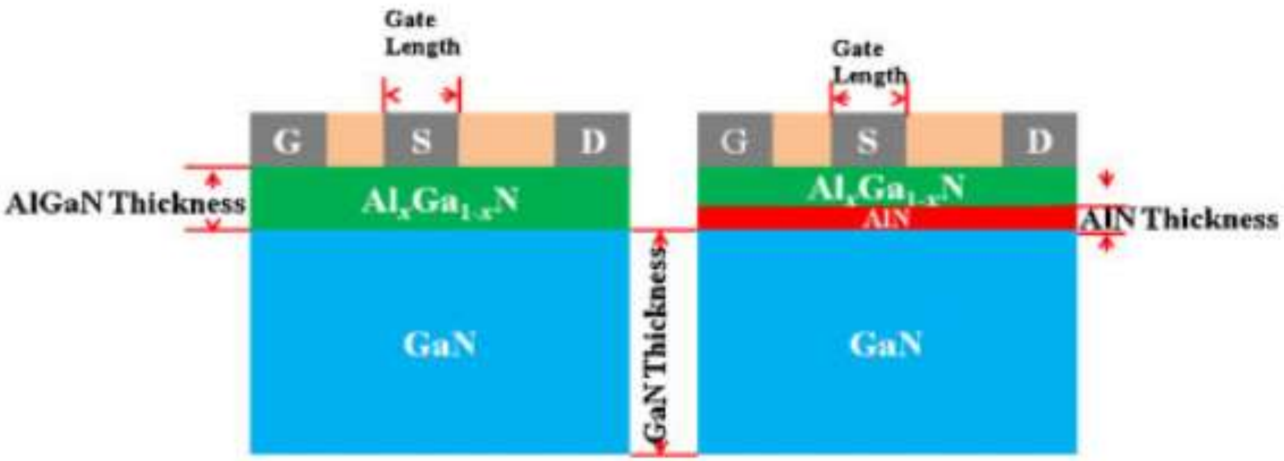
- Epitaxial Device Design
  - Barrier-layer thickness
  - Barrier-layer mole Fraction
  - Cap-layer growth
  - AlN spacer layer ➔ higher mobility
  - Channel concepts & double heterojunction
  - In-situ device passivation, such as SiN_x

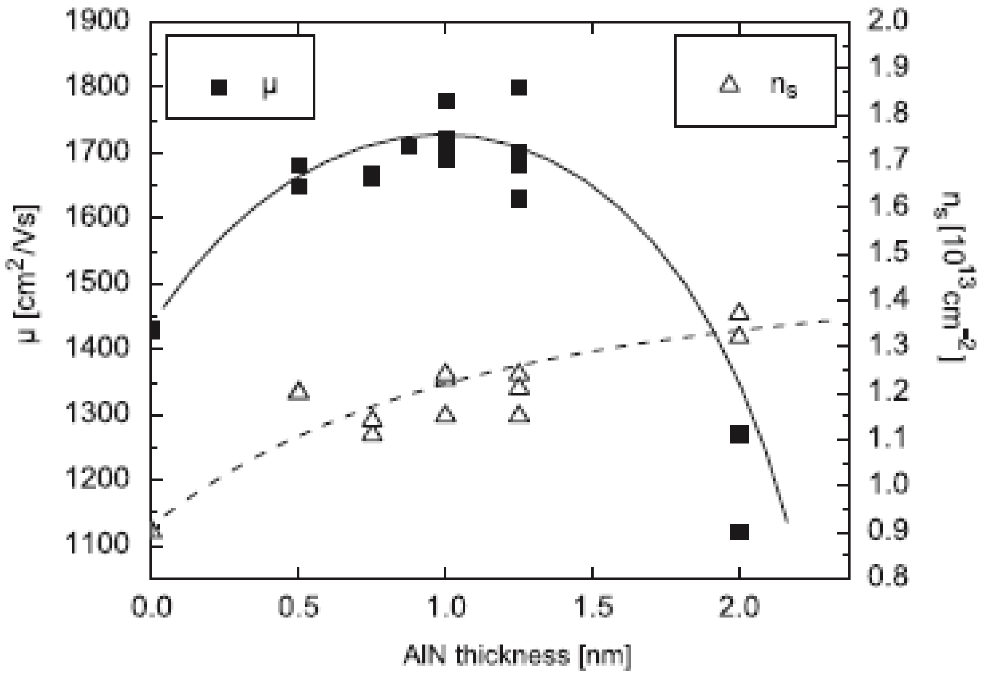


Room temperature 2DEG density measured as a function of $Al_{0.34}Ga_{0.64}N$ barrier thickness

Epitaxy of AlGaN/GaN heterostructure - IV

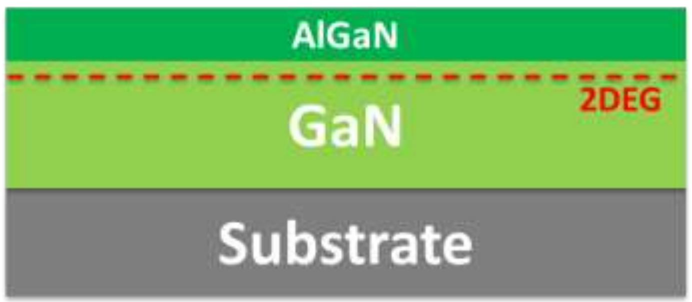
- AlN spacer layer
- Ultra thin layer (<2nm)
- Improve device mobility



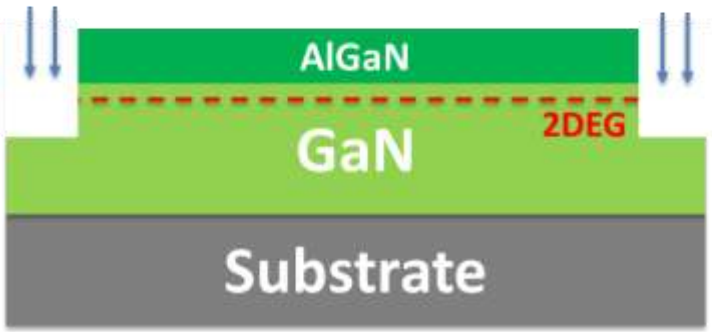


GaN HEMTs fabrication process

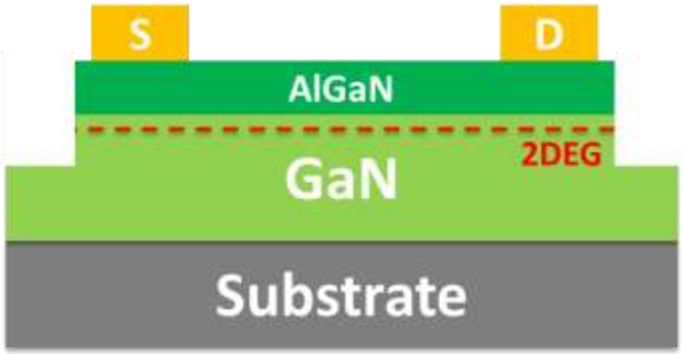
(1) As grown sample



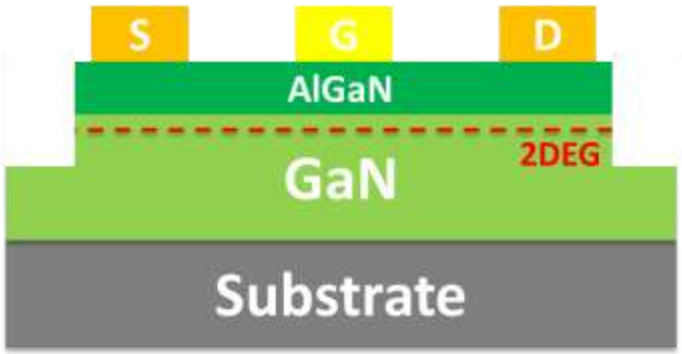
(2) Device isolation



(3) S/D Ohmic contact formation



(4) Schottky gate formation



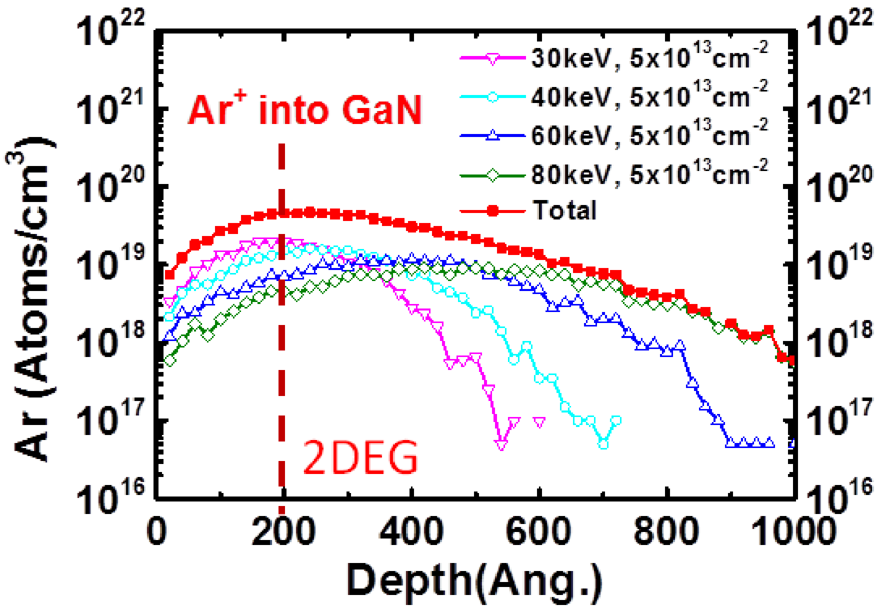
Homework Question: What is the difference between fabrication process of GaN transistors and Si transistors?

Device Isolation

- Mesa etching
  - Simplest technique for electrical isolation
  - Removal of active layers by etching down to an insulating layer or to the substrate
  - Cl_2 based ICP-RIE plasma dry etch
  - Key disadvantages: not a planar device process & etch damage has to be controlled.

Device Isolation

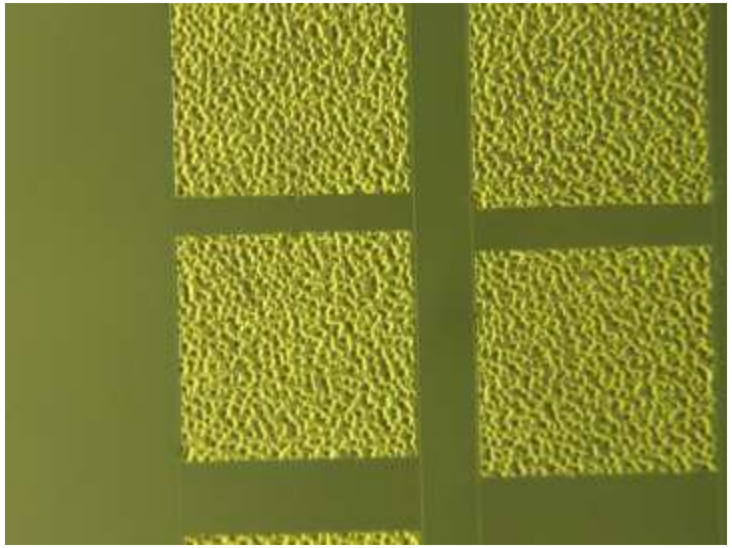
- Implantation
  - Ion implantation processes allow isolation and is a planar device process
  - H, He, F, Ar and N ions
  - Critical parameters: ion mass, ion energy, and implantation temperature
  - Thermal stability is important

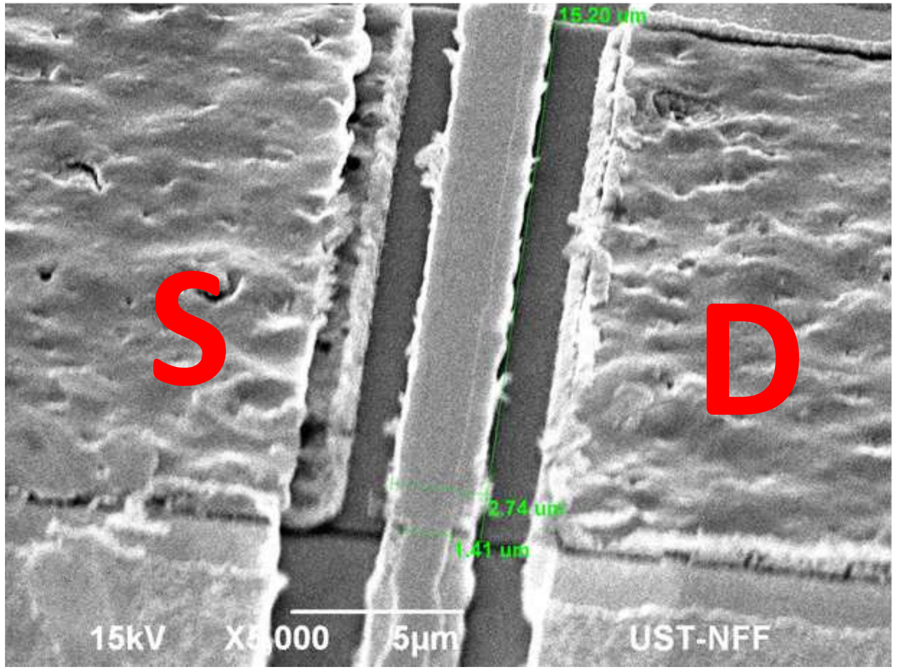


Simulated depth profile of Ar distribution in GaN for device isolation

Ohmic contact -1

- Alloy-contact (between who and who?)
  - Ti/Al/Ni/Au, 850 °C annealing
  - 0.5 – 1 Ω·mm
  - Rough surface and edge

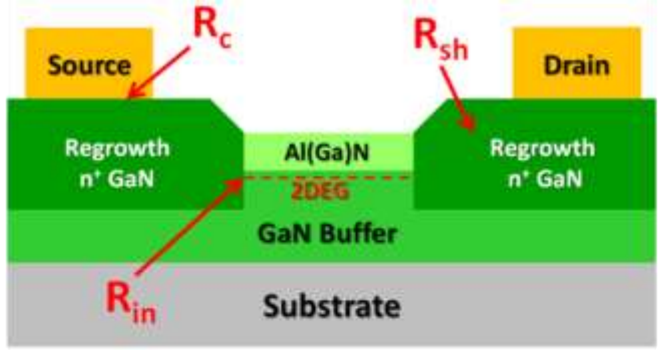


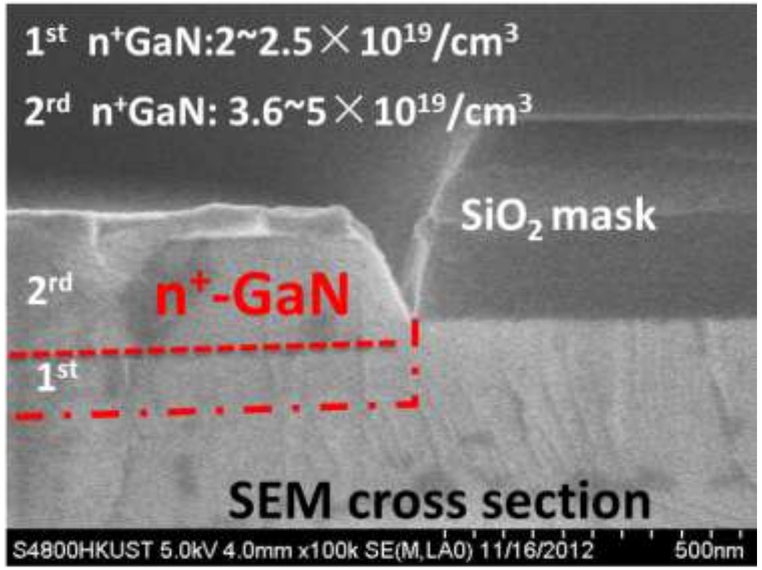


Very rough surface after annealing

Ohmic contact -2

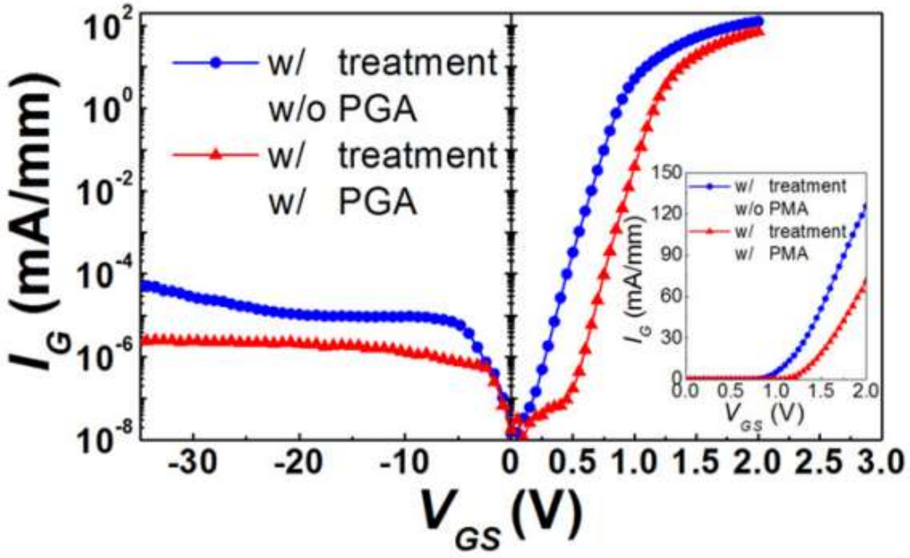
- Regrowth-contact
  - Regrown n^{++}-GaN
  - Growth techinology: Both MBE and MOCVD
  - Resistance as low as 0.1 Ω·mm
  - Mainly for RF application





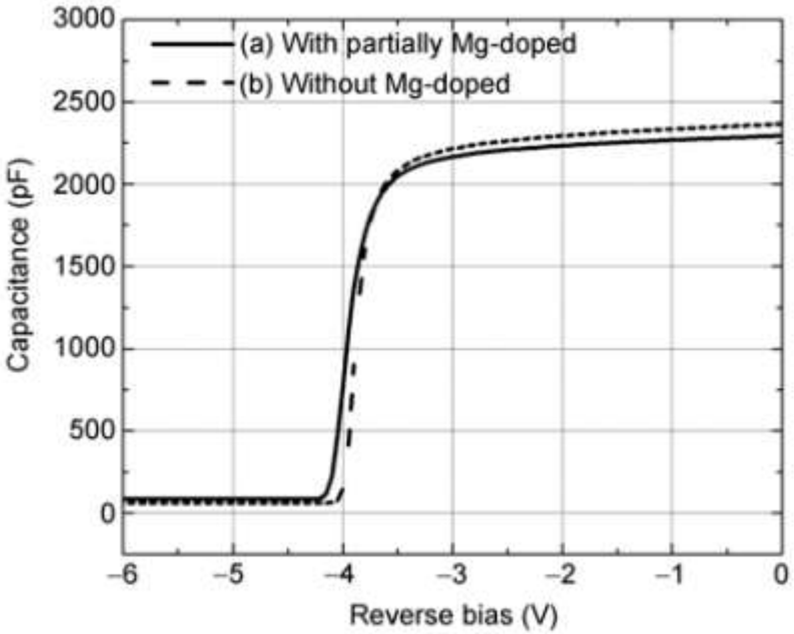
Schottky gate contact

- Schottky-contact properties define charge control and the hot spots of the electric fields in the gate region
- Two important parameters: Schottky barrier Φ_B & ideality factors n
- Typical Schottky metal: Ni or Pt ➔ high Φ_B
- Surface cleaning and pretreatment
- Post gate annealing (PGA)

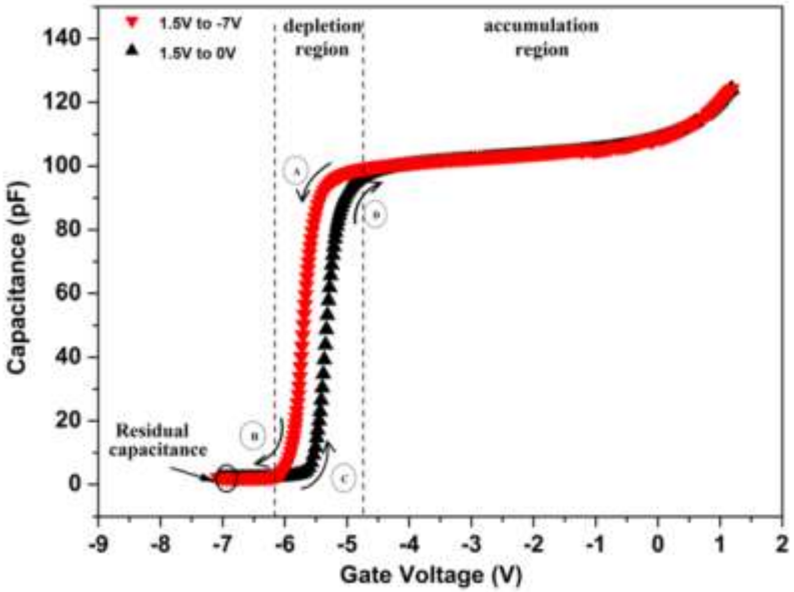


PGA can improve the schottky contact

C-V of GaN HEMT



Mercury-probe C-V profiles for an AlGaN/GaN epi

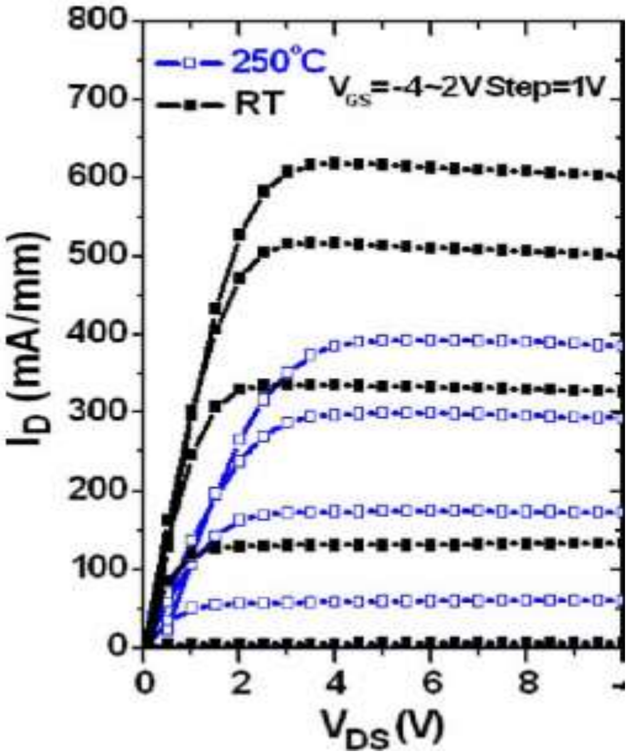


C-V profiles for a Ni/AlGaN/GaN Schottky diode

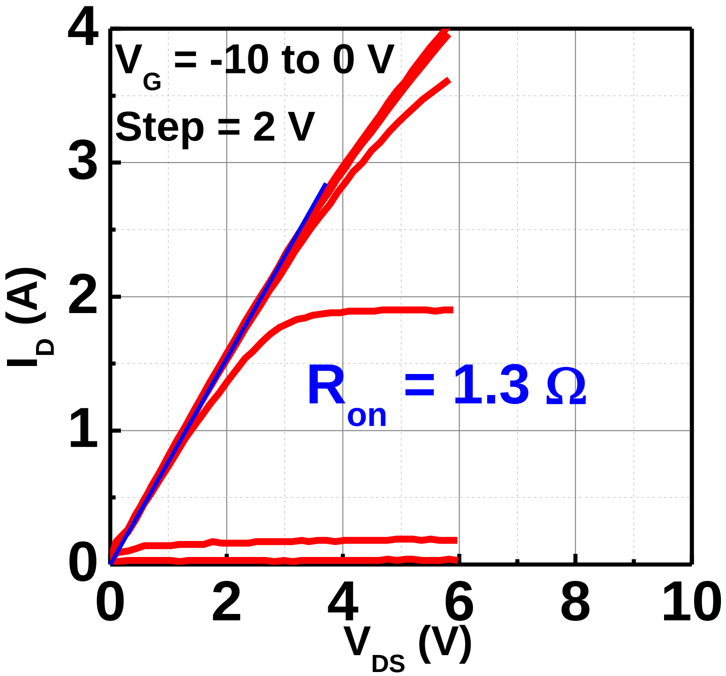
- ✓ C-V properties can be measured before (left figure) and after (Right) device fabrication
- ✓ Threshold voltage shifts negatively after device fabrication
- ✓ Mercury-probe C-V measurement is used for basic characterizations of the material
- ✓ Device C-V reflects more details such as hysteresis

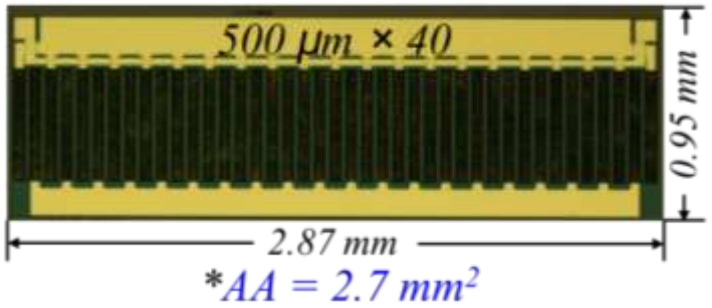
I-V of GaN HEMT

- Output curve



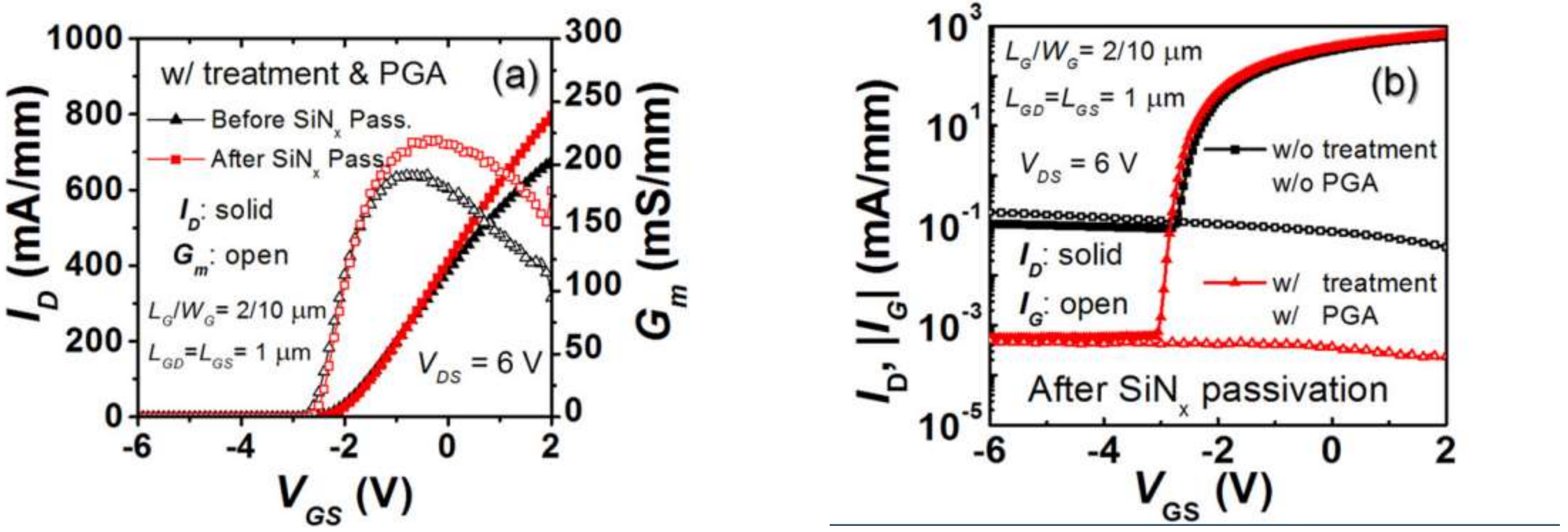
20-mm GaN HEMT



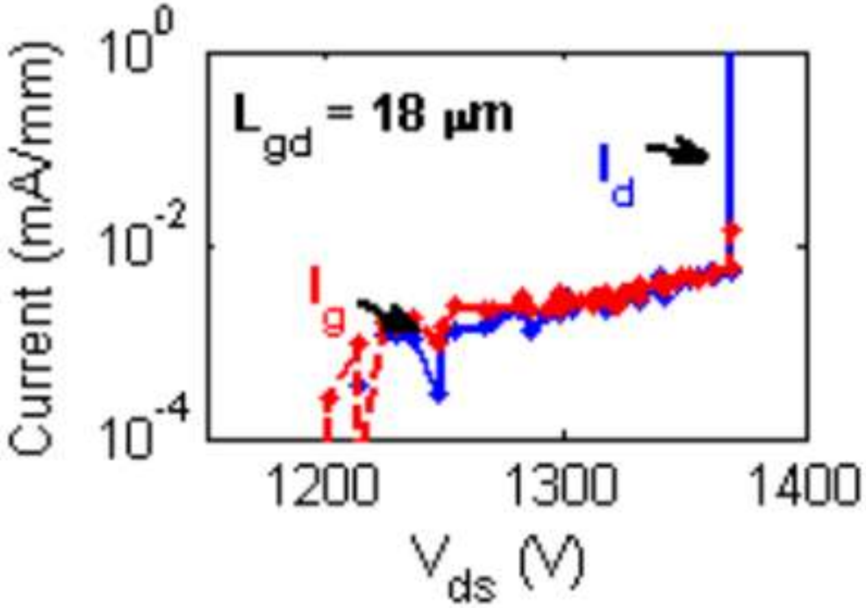


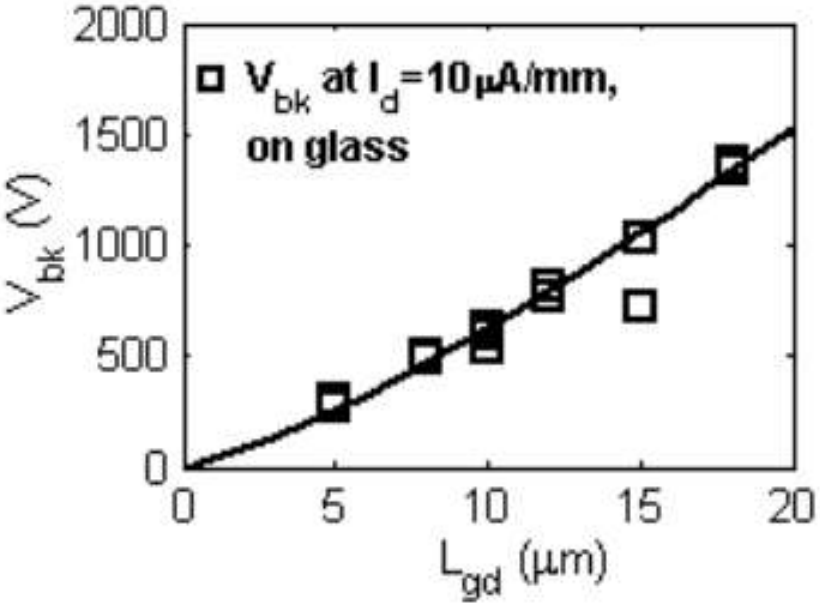
I-V of GaN HEMT

- Transfer curve

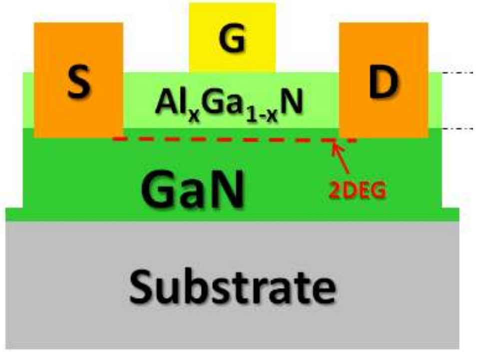


Critical Breakdown Voltage (V_{BD})



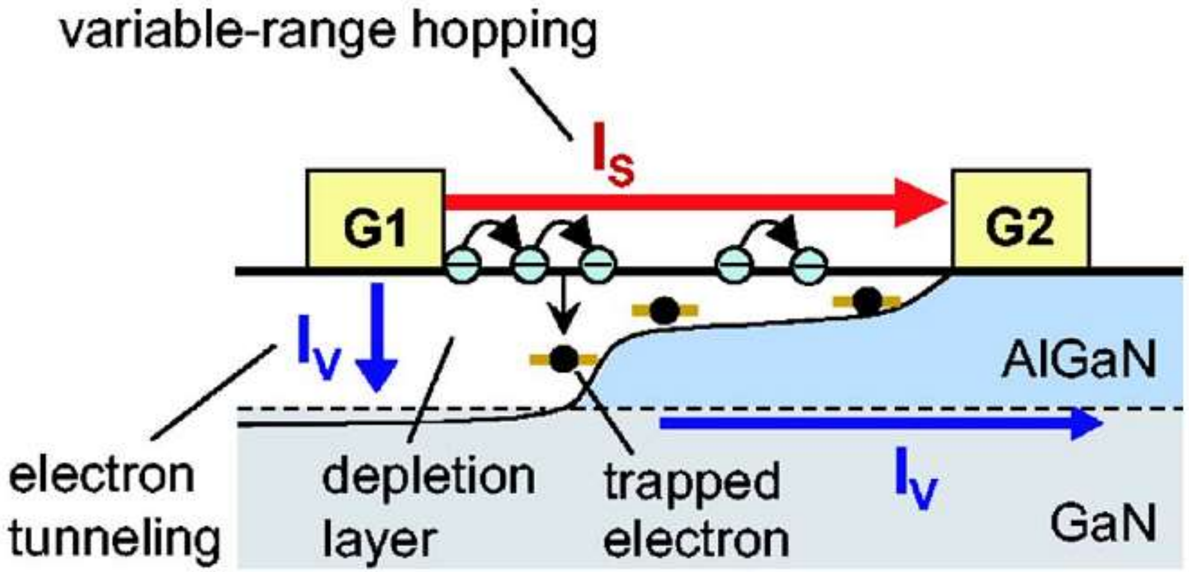


- ✓ Approaches for VBD improvement:
  1. Increase L_{gd}
  2. Leakage current suppression
  3. Improve material (buffer) quality
  4. Field plate design: modify the electric field at the drain side gate edge



Leakage in HEMTs

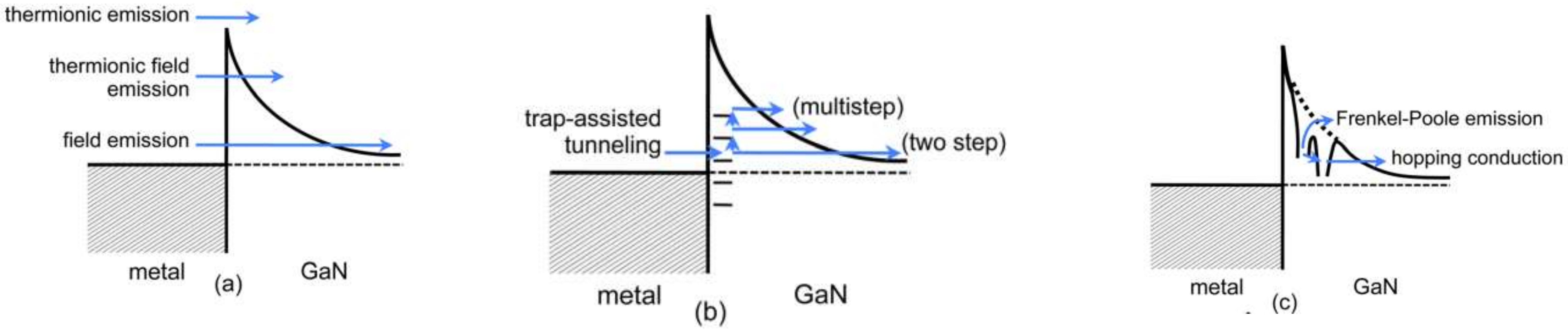
- Leakage Path in an AlGaN/GaN HEMT
  - Electron tunneling leakage
  - Surface leakage



- Solutions:
  - Surface treatment
  - Surface passivation
  - MISHEMTs

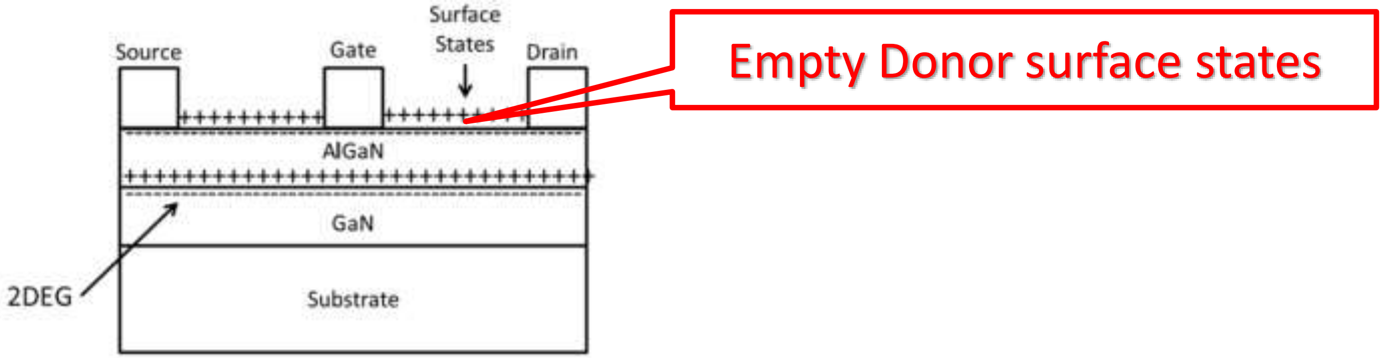
Electron tunneling leakage

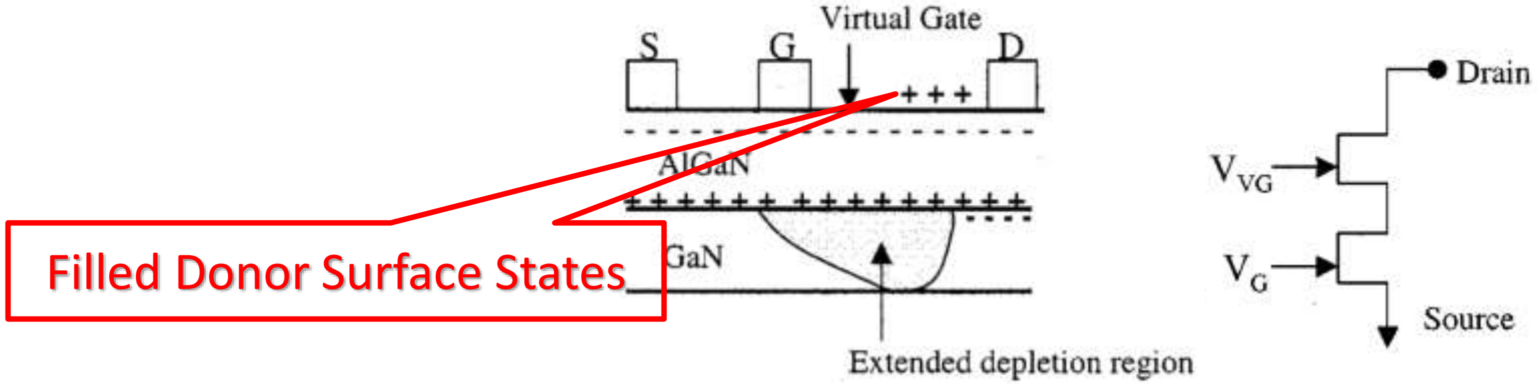
- Tunneling mechanisms in GaN HEMTs



Trapping Effects

- Issues:
  - – Current collapse
  - – Large surface leakage
  - – Low breakdown
  - – V_{th} instability
  - – Poor Reliability
  - – Double gate effect





- Solution: **passivation**

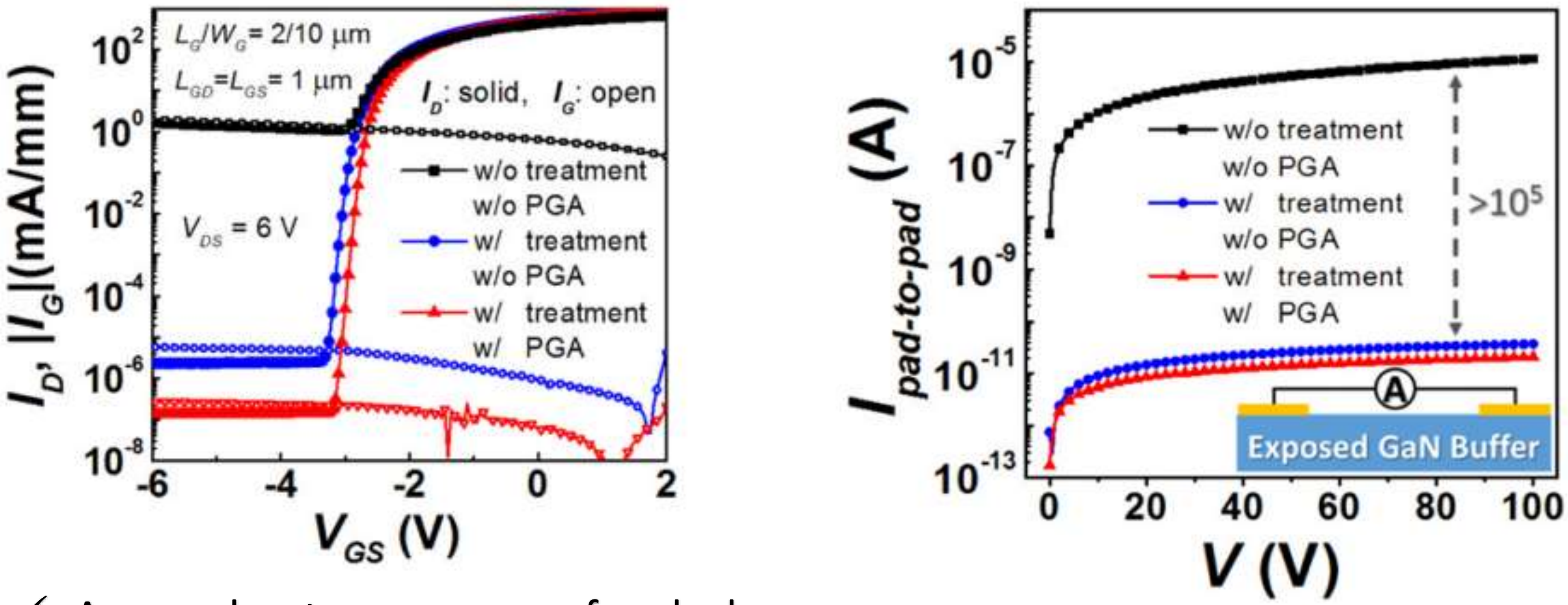
Buffer traps in GaN HEMT

- Under high electric field condition, due to high drain-source voltage, electron moving in the 2DEG channel could get injected into buffer trap
- Due to the longer trapping time constant (about the order of milliseconds) the trapped electrons cannot follow the high frequency signal and hence they are not available for conduction
- The trapped electrons produce negative charge, which deplete the 2DEG, and therefore reduce the channel current
- This greatly affects the Pulse I-V curves of GaN devices

Pulse I-V measurement

- A pulsed I-V measurement is performed by:
- Maintaining a constant bias on the device for most the time
- Periodically applying a short voltage pulse to both gate (V_{gs}) and drain (V_{ds})
- And measuring the drain current
- Commercially available Pulse I-V system will apply pulse of the order of 100’s to milliseconds
- Such pulse lengths are not sufficient for GaN HEMTs
- Increasing the time that the device is under a specific bias however will affect the curves because of self heating
- Nevertheless such effects may be easily isolated and modeled

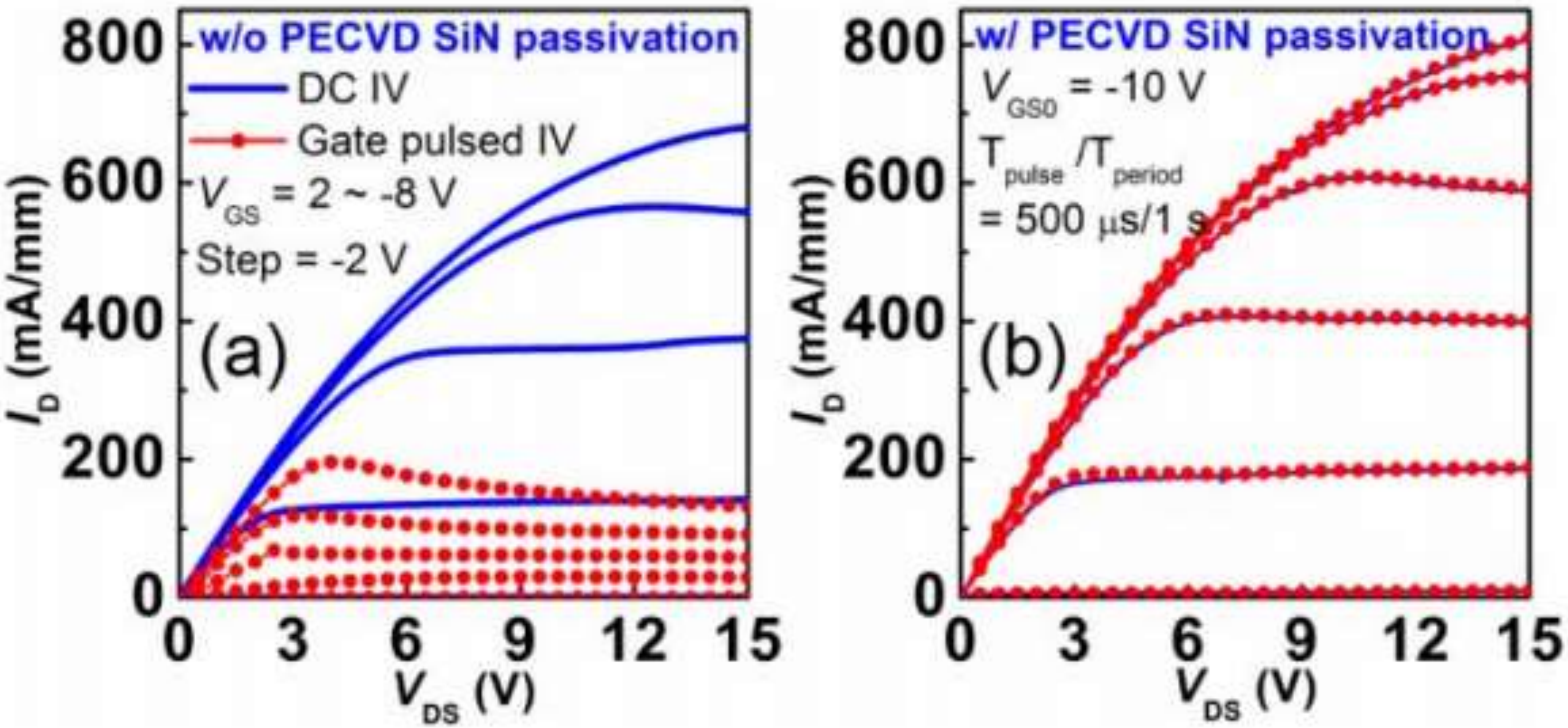
Surface leakage



✓ Approaches to suppress surface leakage:

1. Surface treatment
2. Post gate annealing

Current collapse

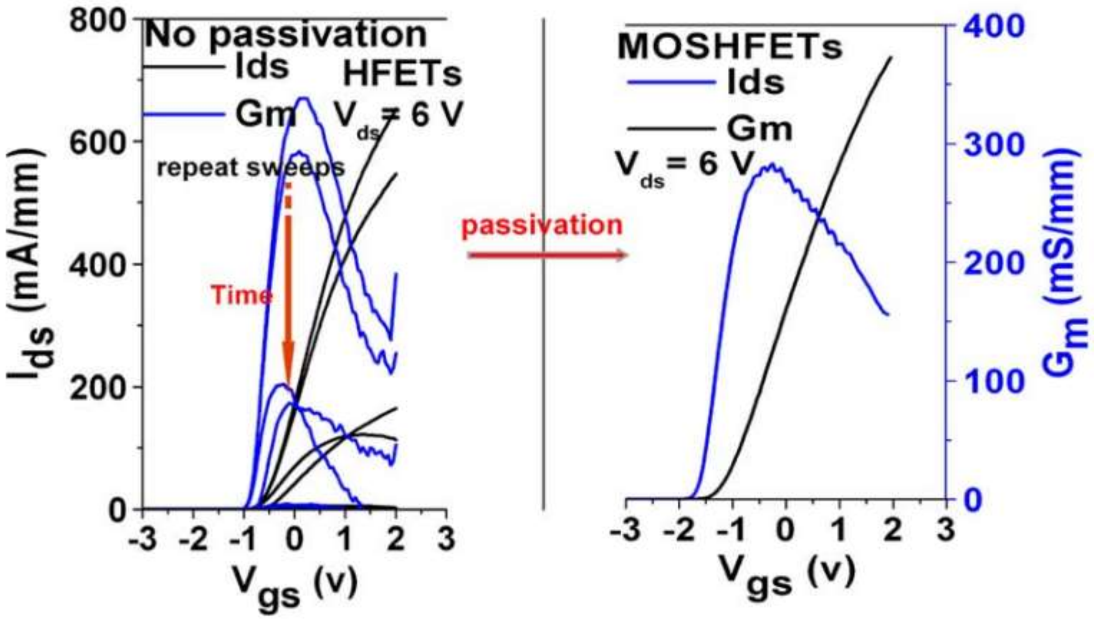


MISFETs or MISHEMTs

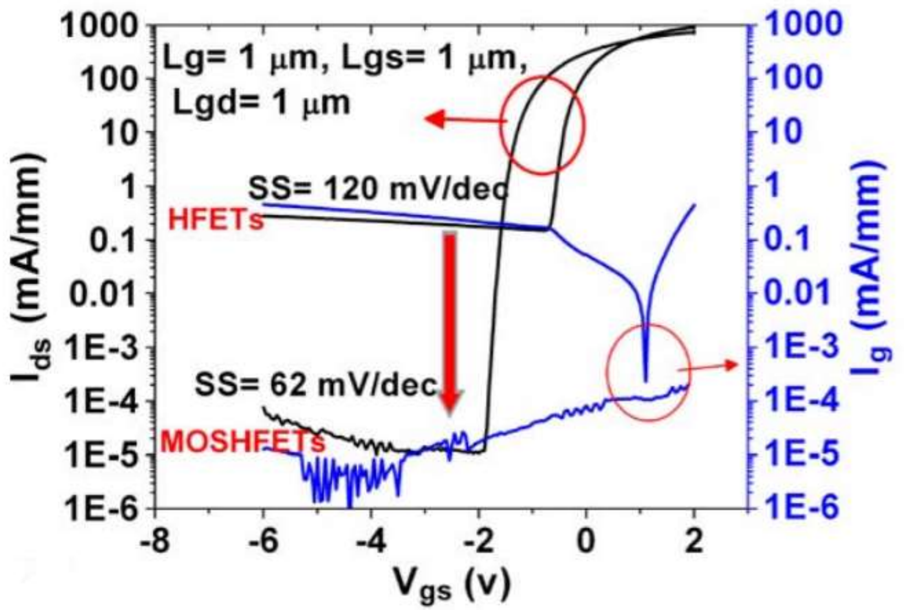
- MISFETs are very attractive candidates for improved performance of the gate diodes relative to nitride HEMTs with metal/semiconductor Schottky gates.
- At the same time MISFETs require the mastery of the insulator/semiconductor interface.
- The material engineering of the semiconductor/dielectric interface of the ungated device region involves not only MIS device but also device passivation

Example-1: AlN MOSHEMT vs. HEMT

- **Surface sensitivity**
  - drain current degradation



- **Large gate leakage**
  - on the order of 1 mA/mm



Passivation-1

- Physically, the deposited dielectric saturates dangling bonds at the ungated semiconductor interface to vacuum. The passivation layer modifies, i.e., mostly reduces the active trap concentration at the interface
- The passivation step is very important, due to the high polarization-induced charges present in III-N devices.

Passivation-2

- Surface passivation has several functions for the device:
  - Protection of surfaces open to the process flow e.g., from the influence of processing steps such as coatings, developer solutions, plasmas, and ambient
  - Removal of surface material and intended damage
  - Modification, typically a reduction of surface states
  - Physical, mechanical, and chemical stabilization of the surface

Process steps

- **Surface cleaning**, i.e.,
  – Material removal
  – Modification of the surface morphology
- **Pretreatment of the semiconductor surface**, i.e.,
  – Chemical reaction at the surface (e.g., oxidation & nitridation)
  – Physical conditioning, (e.g., adsorption of gases)
- **Deposition of a dielectric**, e.g.,
- **Strain engineering by subsequent layers**

Surface clean & pretreatment

- Both the GaN and AlGaN surfaces are found to be highly corrosive when exposed to air or other ambient, so the exposure of the surface to air and any kind of processing medium is found to be critical
- During device fabrication, surface treatment, and cleaning conditions are very important and at the same time unavoidable

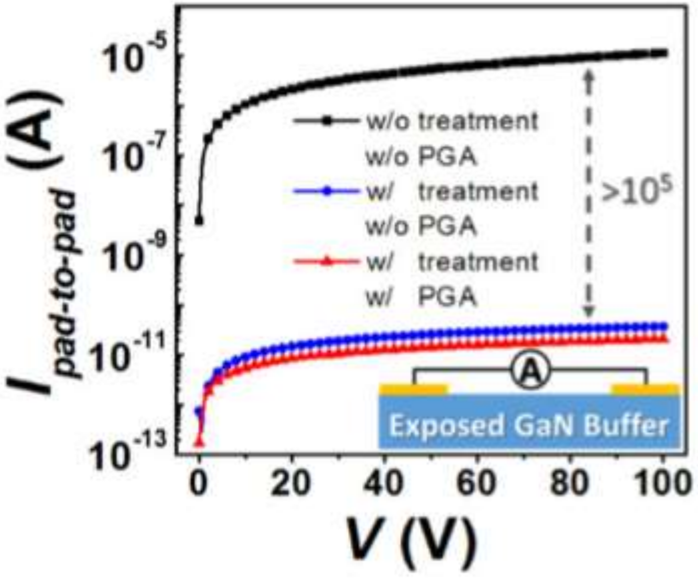
Surface clean Procedures

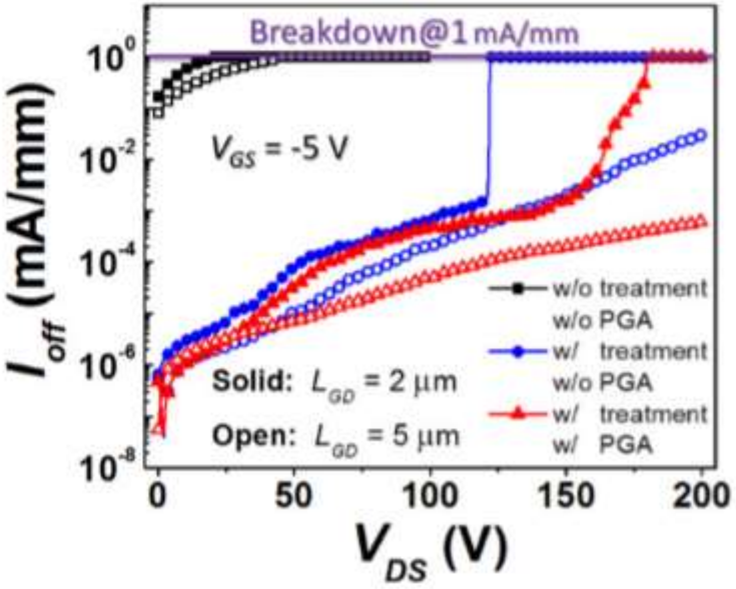
- **Surface clean process:**

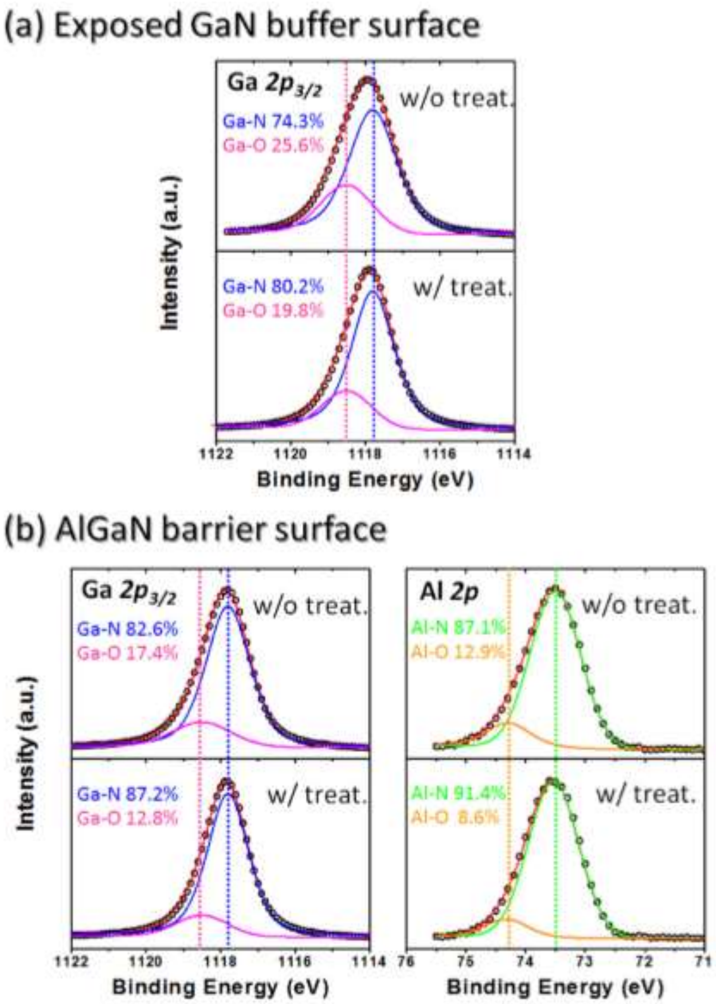
1. Acetone, isopropanol (IPA), and DI water
2. Buffered oxide-etch (BOE) or HCl dip
3. Other solutions, such as NH_4OH, KOH, H_2SO_4, TMAH, etc.

Pretreatment examples-1

- Two step process: a low power O_2-plasma and HCl surface treatment



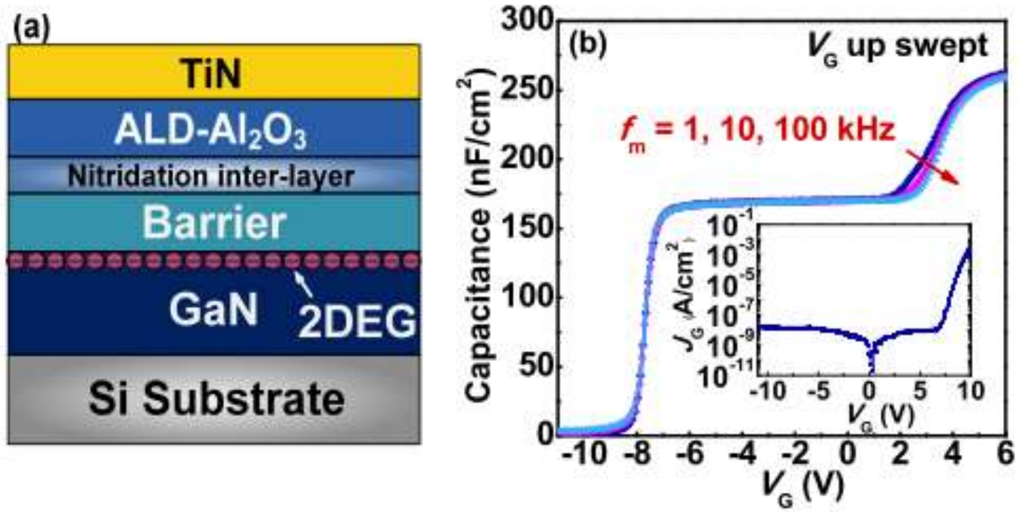




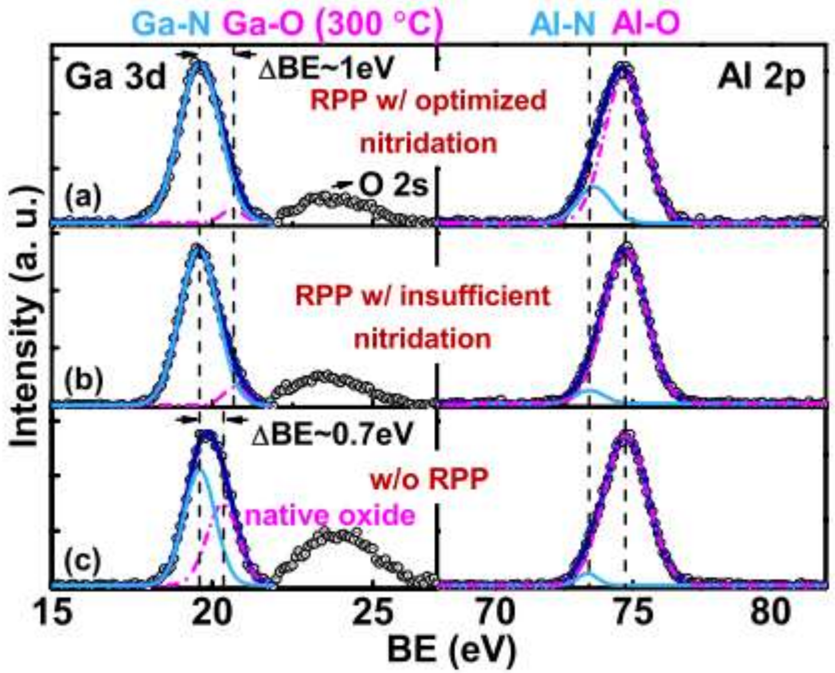
Reduction in off-state leakage current, mainly due to the removal of native oxide from the surface (as evidenced by XPS).

Pretreatment examples-2

- Surface nitridation



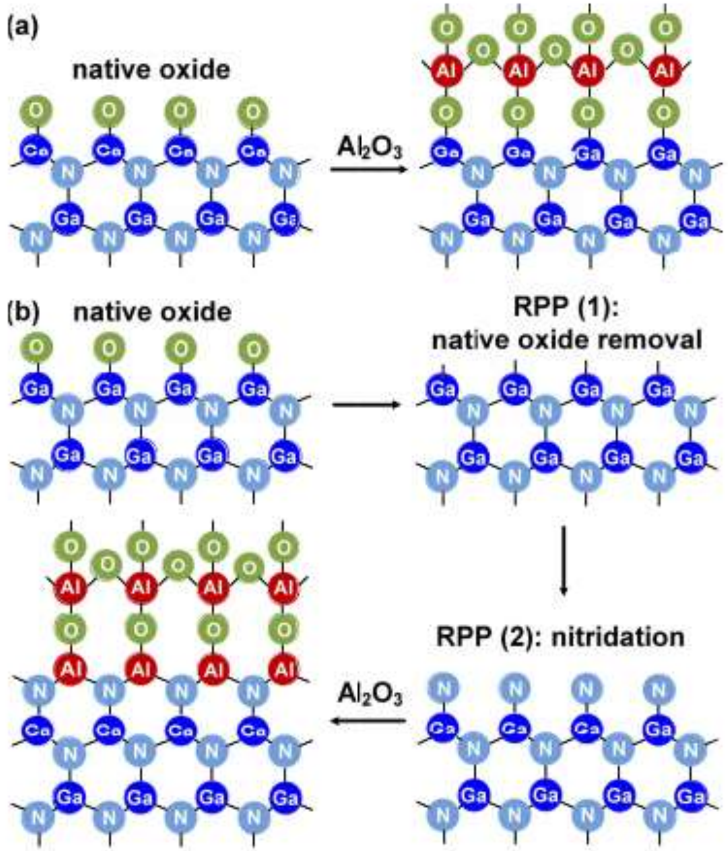
Surface nitridation prior to the gate dielectric deposition



Ga 3d and Al 2p core-level spectra of the three samples with and without nitridation in XPS analysis; Nitridation can effectively remove the native oxide on the GaN surface.

Pretreatment examples-2

- Surface nitridation



key mechanism of the in situ pre-gate treatment in improving the Al_2O_3/GaN-cap interface

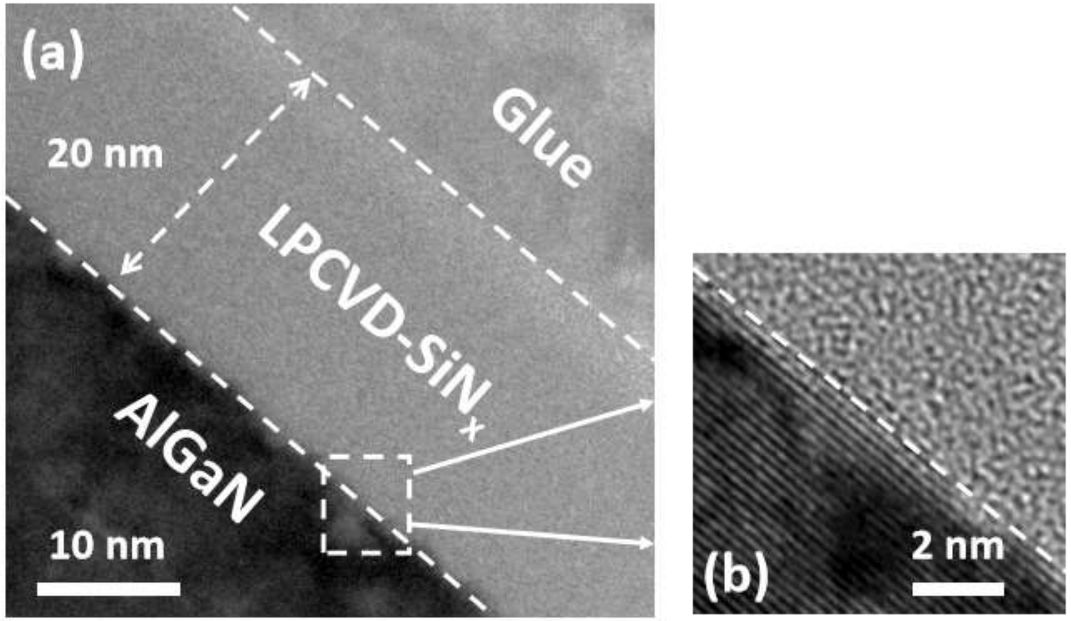
First, in situ RPP (Remote Plasma Pretreatment) with NH_3-Ar plasma can effectively remove the amorphous native oxide on the surface. Second, with optimized in situ plasma nitridation, a thin layer of monocrystal-like NIL (containing Al-N bonds) is formed prior to the subsequent Al_2O_3 deposition. The NIL acts as an oxygen barrier to suppress the Ga-O bond formation during the subsequent Al_2O_3 deposition, resulting in a uniform and sharp interface between Al_2O_3 and GaN-cap.

Dielectric deposition

- Silicon Nitrides (Si_3N_4)
  - Plasma-enhanced chemical vapor deposition (PECVD)
  - Low pressure chemical vapor deposition (LPCVD)
  - Catalytic chemical vapor deposition (CAT-CVD)
  - MOCVD in-situ epitaxy
- Al_2O_3 by ALD
- High-k insulators: HfO_2, ZrO_2, etc.

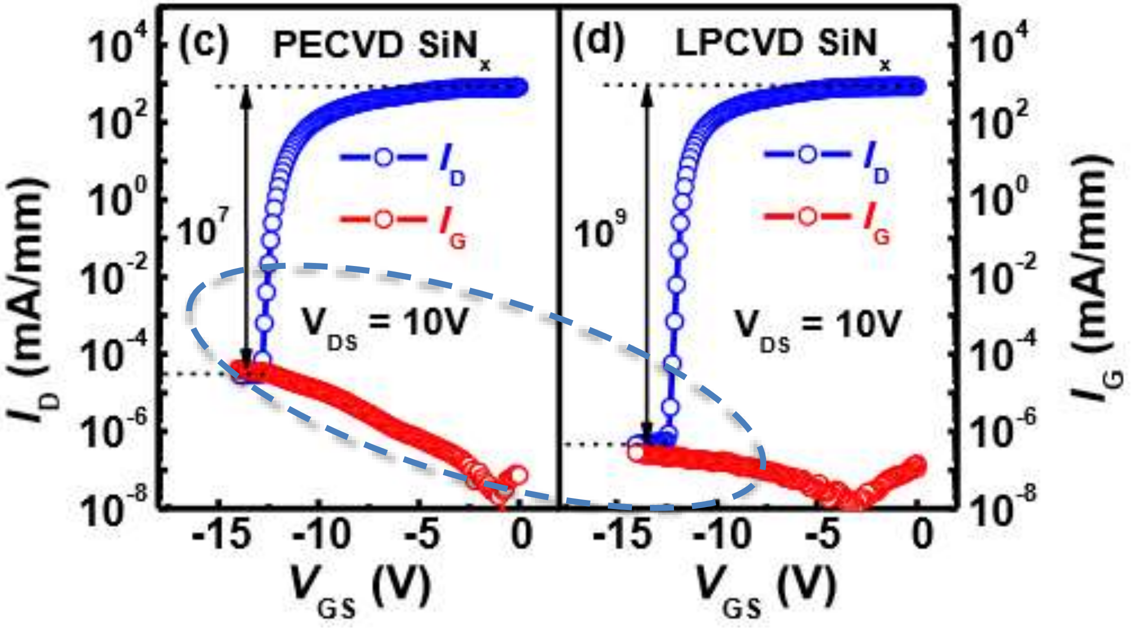
LPCVD vs. PECVD SiNx MISHEMT-1

- LPCVD advantages:
  - free of plasma-induced damage
  - higher film quality including larger breakdown electric field, higher dielectric constant and higher thermal stability, due to the high growth temperature (> 600 C)

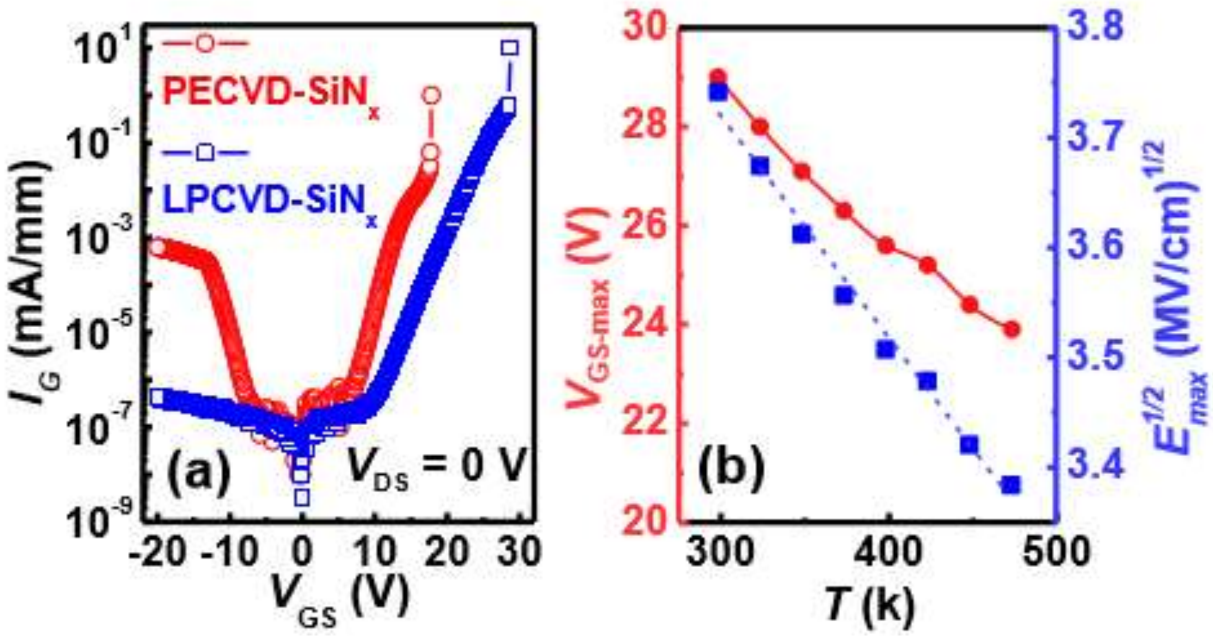


Cross-sectional TEM micrograph of the LPCVD-SiNx/AlGaN stack with very sharp dielectric/semiconductor interface.

LPCVD vs. PECVD SiNx MISHEMT-2

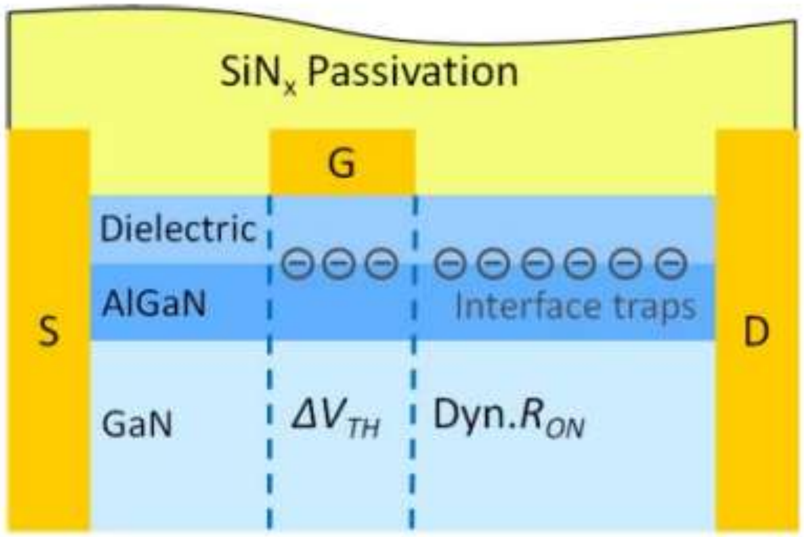


Transfer and gate leakage characteristics of MIS-HEMTs with PECVD SiNx and LPCVD SiNx gate dielectric.
Device dimensions: $L_G/L_{GS}/L_{GD}$ = 1.5 /2 /15 μm.



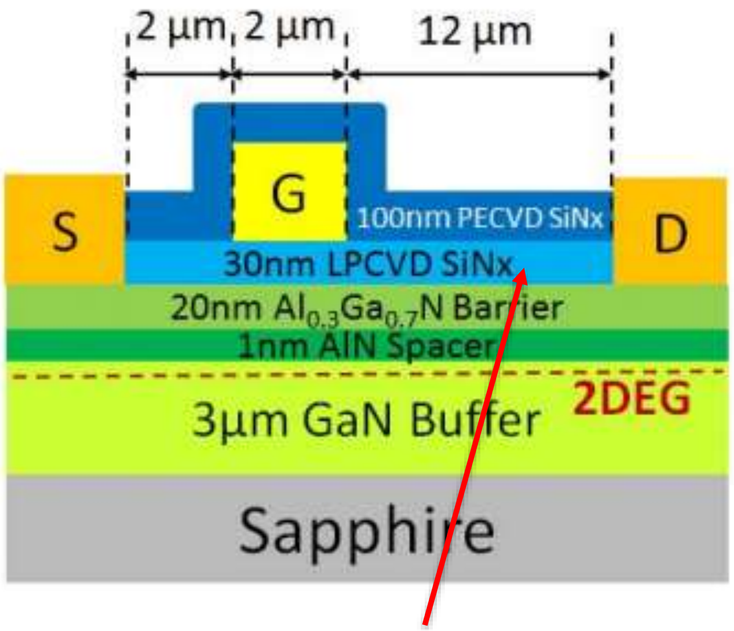
I_G–V_G characteristics of the LPCVD-SiNx/AlGaN/GaN and PECVD-SiNx/AlGaN/GaN MIS-HEMTs; Temperature dependent forward gate breakdown characteristics of LPCVD-SiNx MIS-HEMTs.

Bilayer SiNx passivation-1



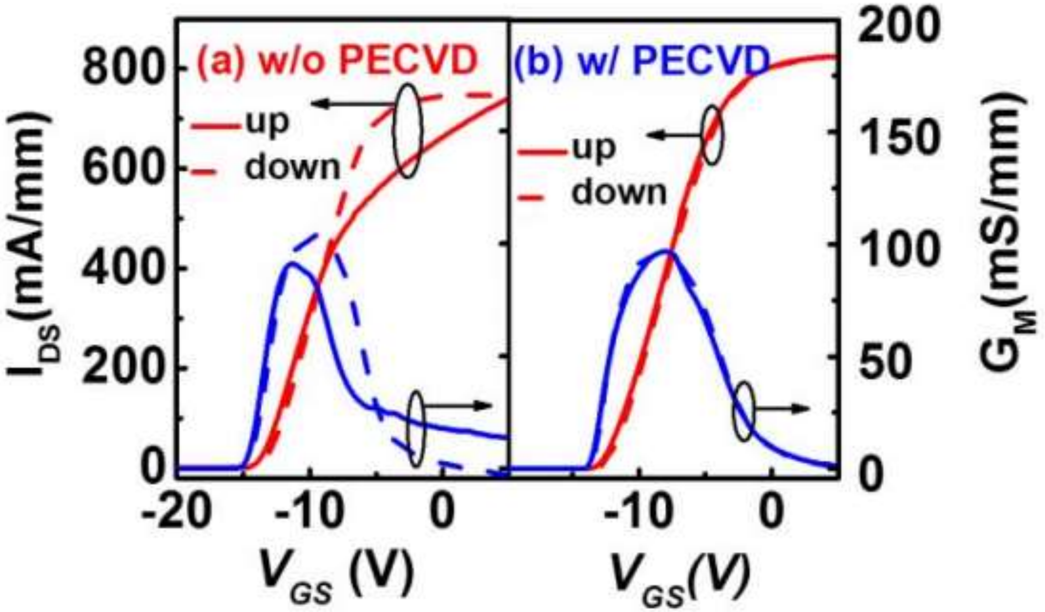
Interface trap locations and their influence on electrical performance in AlGaN/GaN MISHEMTs: V_{th} instability & dynamic R_{on} increases

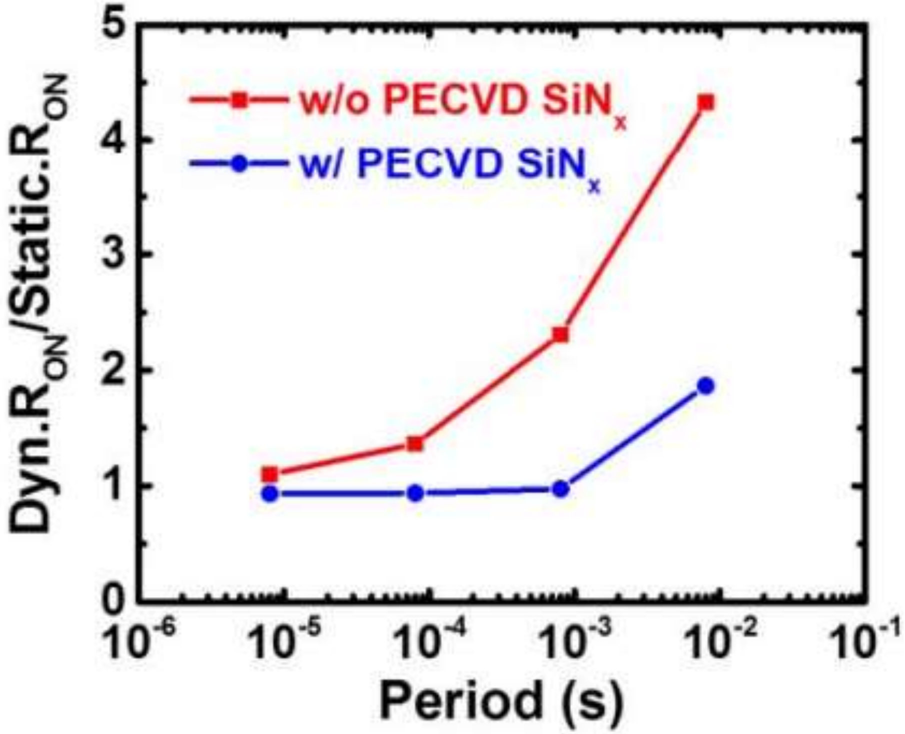
Schematic diagrams of the LPCVD SiNx /AlGaN/GaN MISHEMT with PECVD SiNx passivation.



Bilayer SiNx passivation in access region

Bilayer SiNx passivation-2





Bilayer passivation

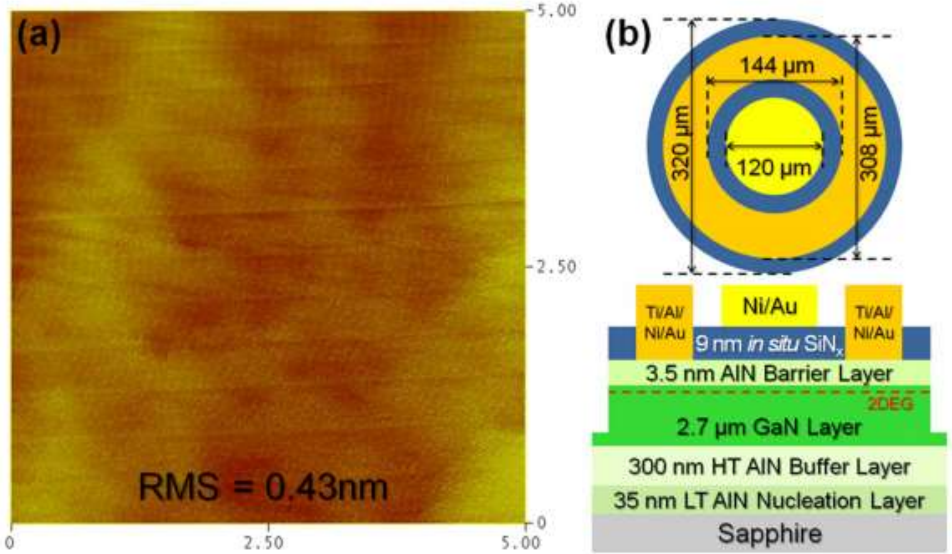
✓ Small hysteresis in double-mode measurement

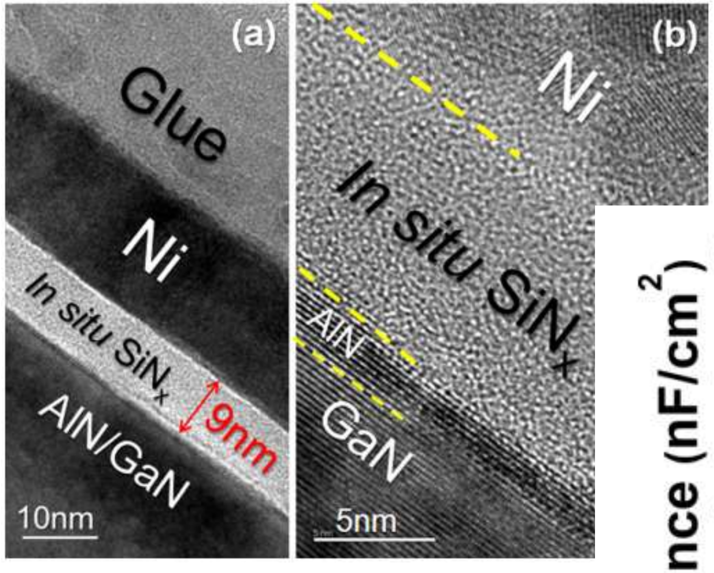
✓ Small dynamic Ron increases

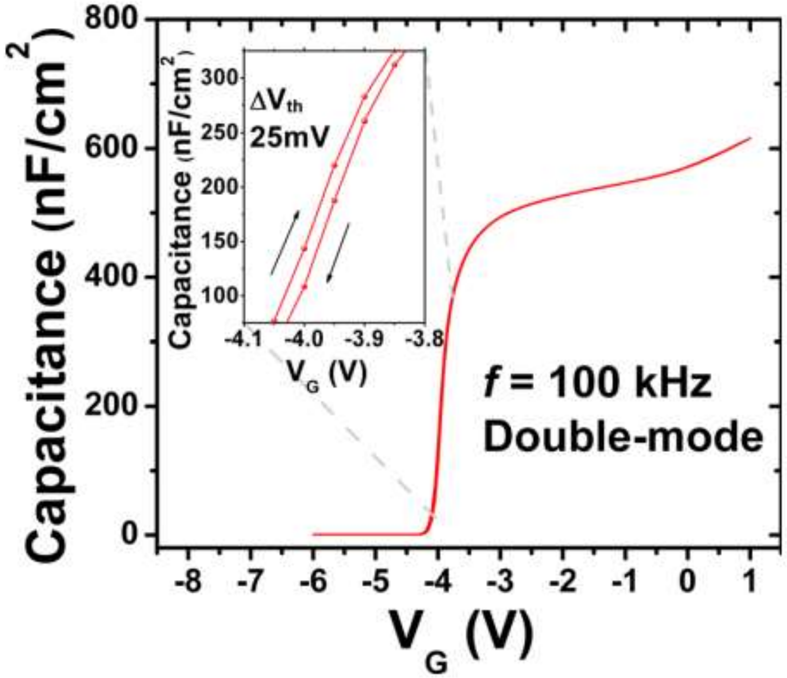
MOCVD in-situ SiNx

- Advantages:
  - An in situ SiNx gate dielectric can be epitaxially grown immediately after the GaN HEMT barrier growth in a MOCVD system
  - Without the expose of the as-grown GaN HEMT barrier to the ambient air, the process-induced contamination or damage can be prevented.
  - Moreover, the low growth rate and high growth temperature of in situ SiN are favorable for good dielectric quality.

In situ SiNx/AlN/GaN MIS



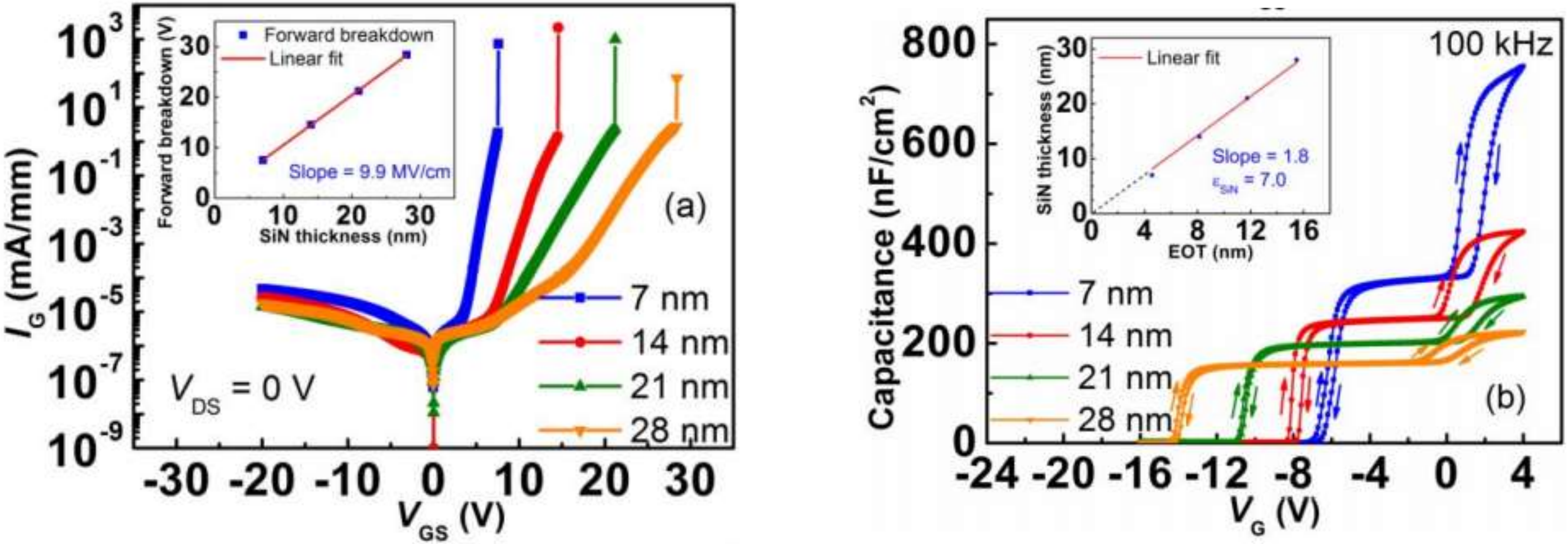




The in situ SiNx was deposited at 1145 ℃ by MOCVD using silane and ammonia as precursors.

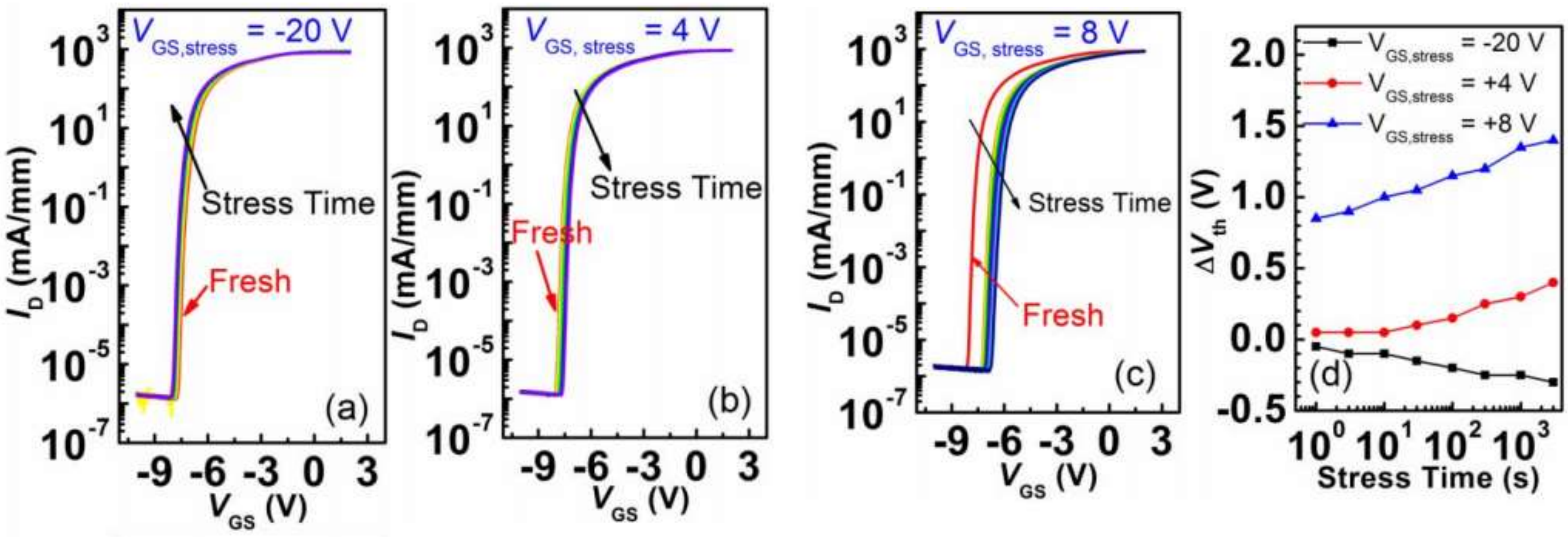
Good interface quality with small hysteresis in the C-V curve

In situ SiNx MISHEMT



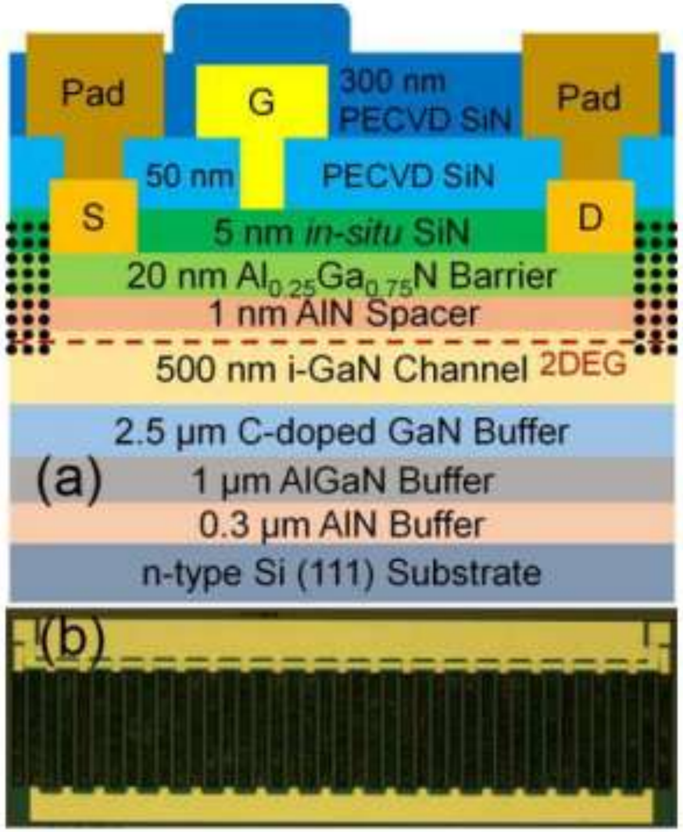
Forward and reverse gate leakage of GaN MISHEMTs and (b) double sweep C–V characteristics of circular MISCAPs with various in situ SiNx thicknesses. Inset in (a): forward gate breakdown voltage dependence on the SiNx thickness. Inset in (b): in situ SiN thickness versus the corresponding EOT.

V_{th} instability

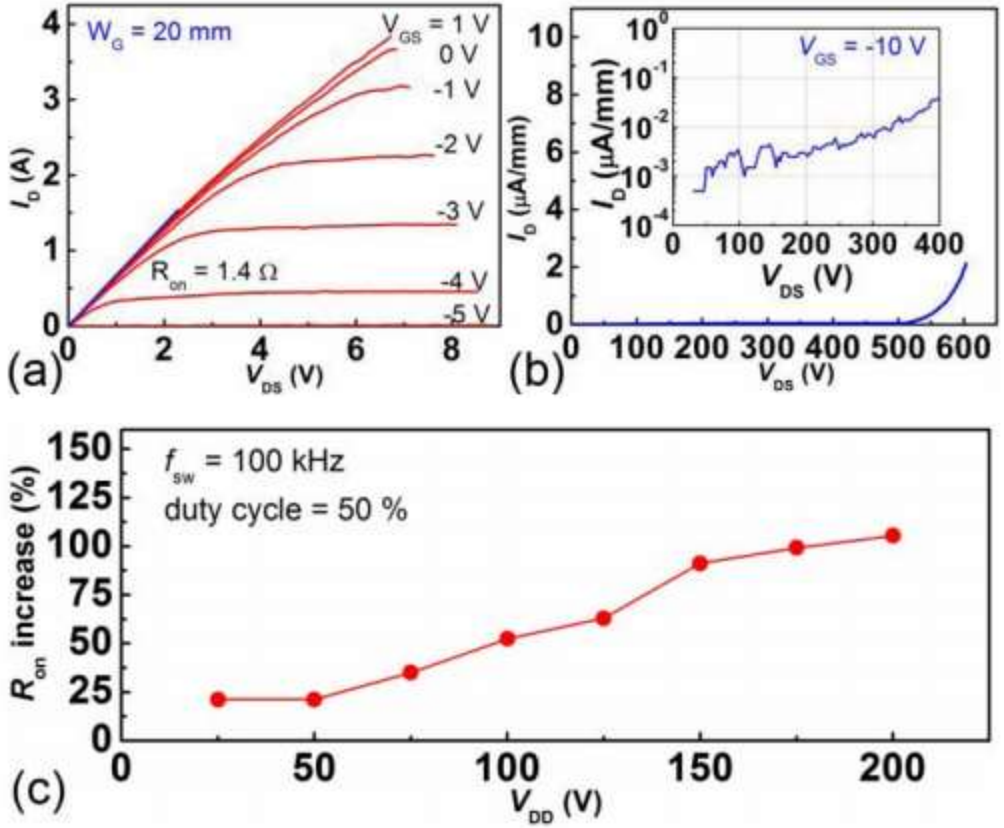


I_D–V_{GS} curves measured during the 3000 s. (a) Reverse gate bias stress (V_{GS} = − 20 V). (b) Forward gate bias stress of V_{GS} = 4 V. (c) Forward gate bias stress of V_{GS} = 8 V. (d) Threshold voltage shift ($\Delta V_{th} = Vth_{stress} - Vth_0$) during the stress in (a)–(c).

Large gate width device

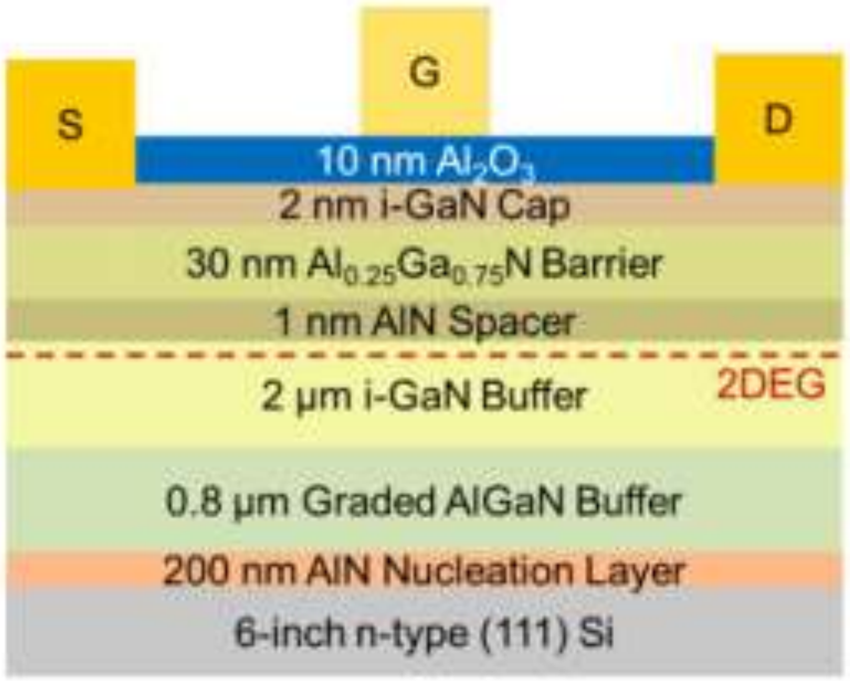


(a) Cross-sectional schematic of GaN power MISHEMTs on Si.
(b) Overview of the MISHEMTs with a gate width of 20 mm.

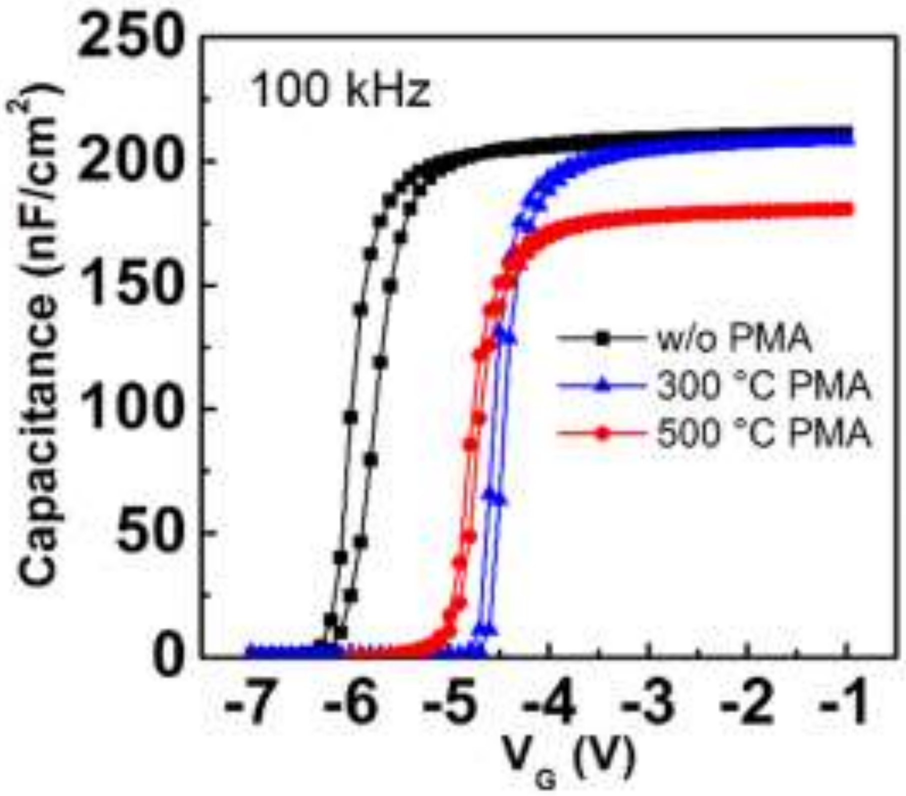


Increase of the dynamic R_{ON} over the static R_{ON} versus VDD

ALD Al_2O_3 MISHEMT-1

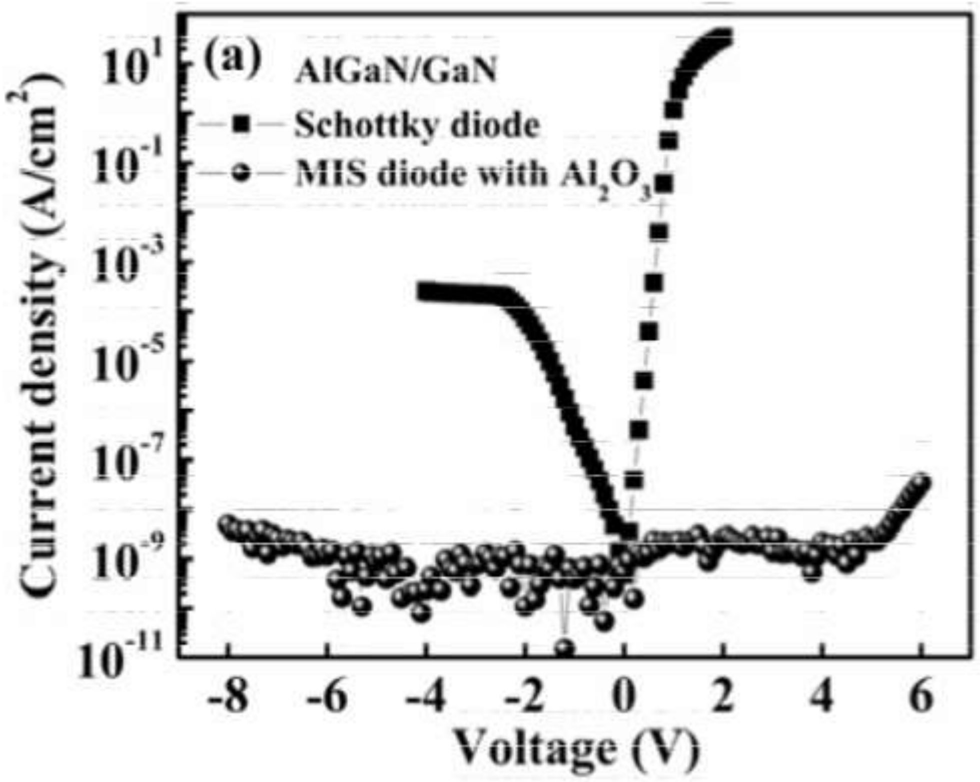


Cross-sectional schematic of a MOSHEMT

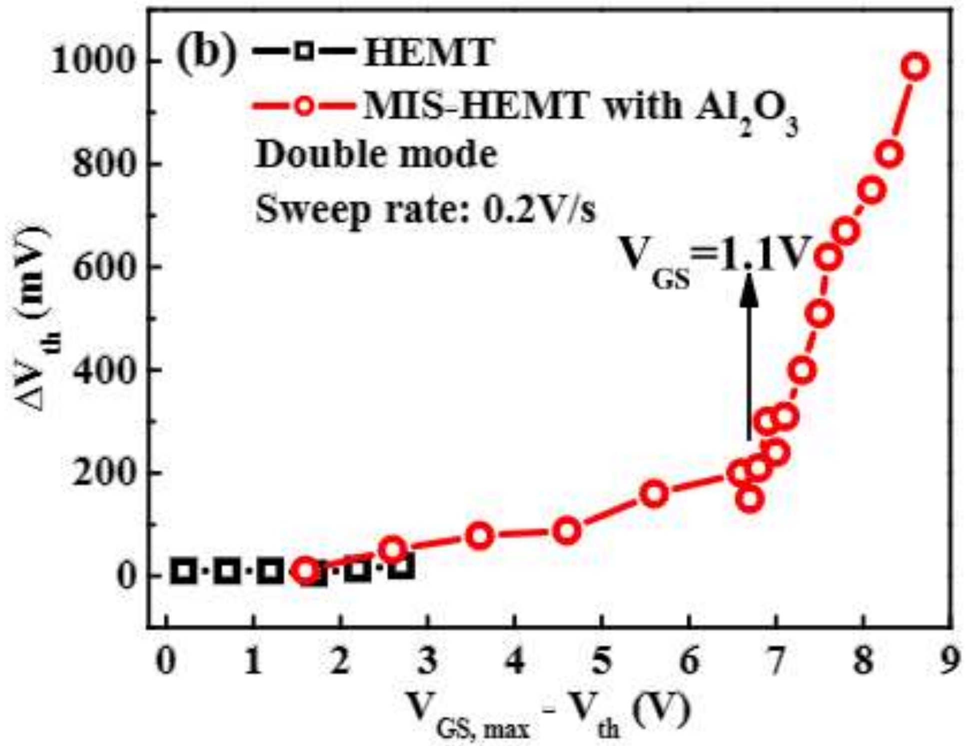


Capacitance–voltage characteristics of circular MOSCAPs without and with various PMAs; V_{th} shifts towards positive after PMA.

ALD Al_2O_3 MISHEMT-2



Reduced gate leakage current by ALD Al_2O_3



Vth instability issues: evolution of ΔV_{th} with $V_{GS,max}$-V_{th} in HEMTs and MIS-HEMTs

RF Transistors-1

- The current gain cut-off frequency (f_T) and the maximum oscillation frequency (f_{max}) are two figures of merit (FoMs) for the high frequency performance of a transistor
- f_T is the frequency at which the short-circuit current gain of the device decreases to unity
- f_{max} is the highest frequency at which power gain can be obtained from a device.

RF Transistors-2

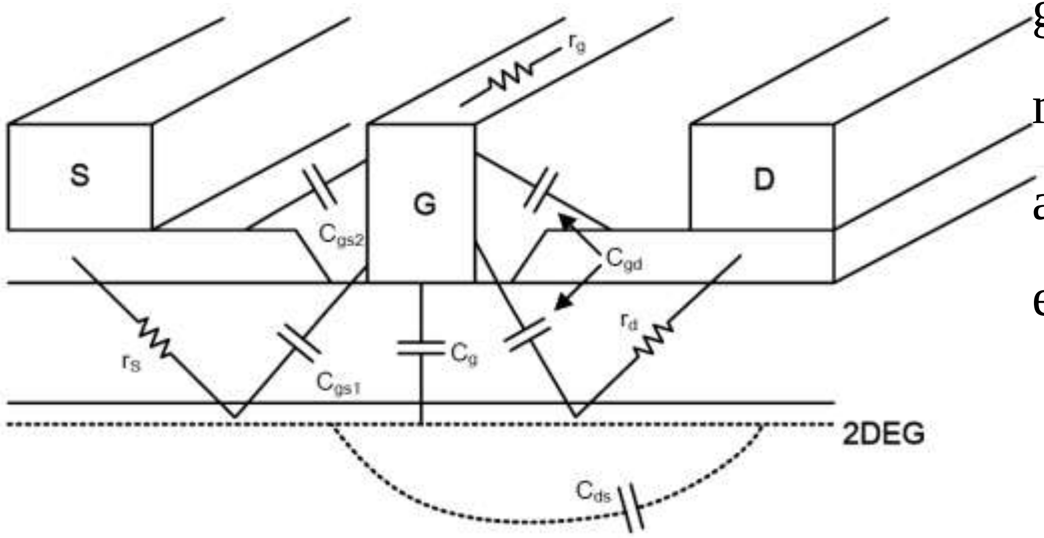
- Considering the physical mechanism of a transistor operation, f_T can be represented by the channel electron drift velocity through the following equation:

$$f_T = \frac{v_{sat}}{2\pi L_G}$$

- where v_{sat} is the saturation electron drift velocity and L_G is the gate (channel) length (L_G). It is apparent that a higher electron saturation velocity and smaller L_G in higher f_T.

GaN HEMT for RF-1

- The physical origins of the elements in an equivalent circuit model of a GaN HEMT



The g_m is the measure of the devices' intrinsic gain, representing the effectiveness of the gate in modulating the drain current. When taking into account the parasitic components, such as R_s, the extrinsic transconductance (G_m) can be described as

$$G_m = \frac{g_m}{1 + g_m \cdot R_s}$$

GaN HEMT for RF-2

- f_T is the frequency at which the short circuit current gain ($|h_{21}|^2$) of the device falls to unity. In the first order approximation, the equation below gives the definition of f_T:

$$f_T = \frac{g_m}{2\pi(C_{gs}+C_{gd})}$$

GaN HEMT for RF-3

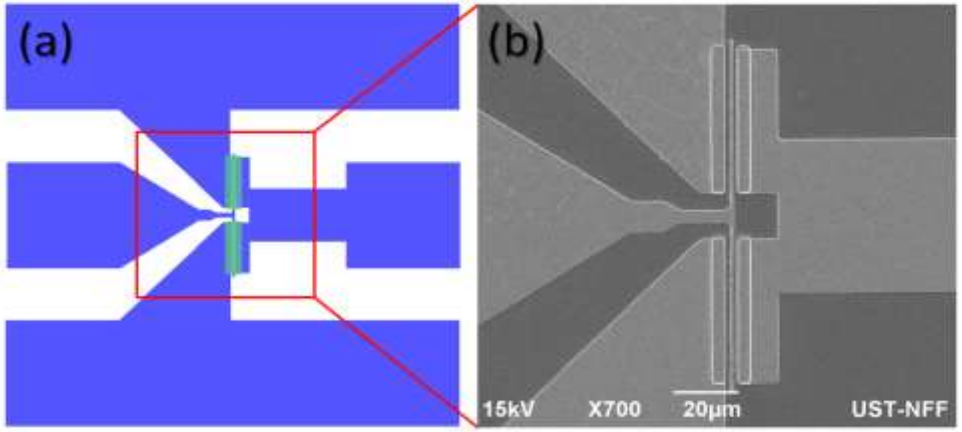
- f_{max} is used as an indicator of the ultimate frequency limits of a device, which is defined as the frequency at which the power gain of a device reaches unity

$$f_{max} = \frac{f_T}{2}\sqrt{\frac{R_{ds}}{R_g + R_{in}}}$$

- To maximize the f_{max}, the f_T and the resistance ratio $\frac{R_{ds}}{R_g + R_{in}}$ of the HEMT have to be optimized.

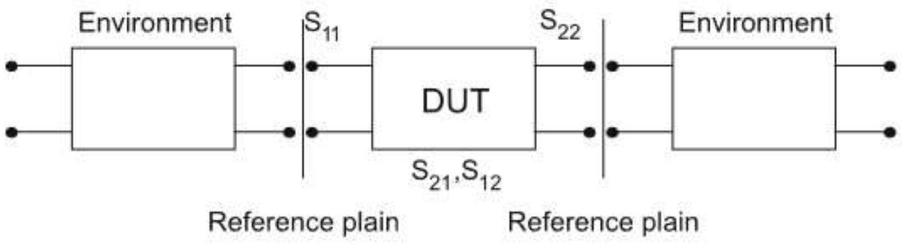
RF characterization & small signal modeling-1

- On-wafer S-parameter measurements
  - GSG RF probe & network analyzer



The device layout and a top-view SEM image of the gate region

S-parameter measurements

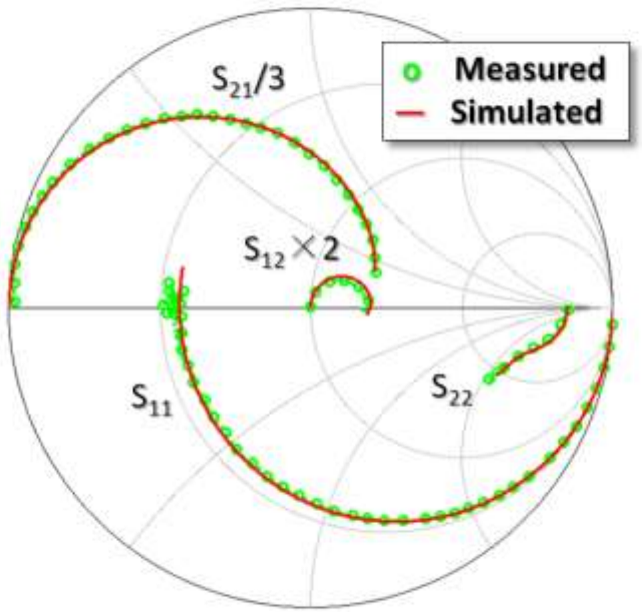


Derived from S-parameter measurements at a given impedance, e.g., 50Ω, all other high-frequency parameters, such as h-, Z-, Y-, and a-parameters, are available and can be converted to each other unambiguously.

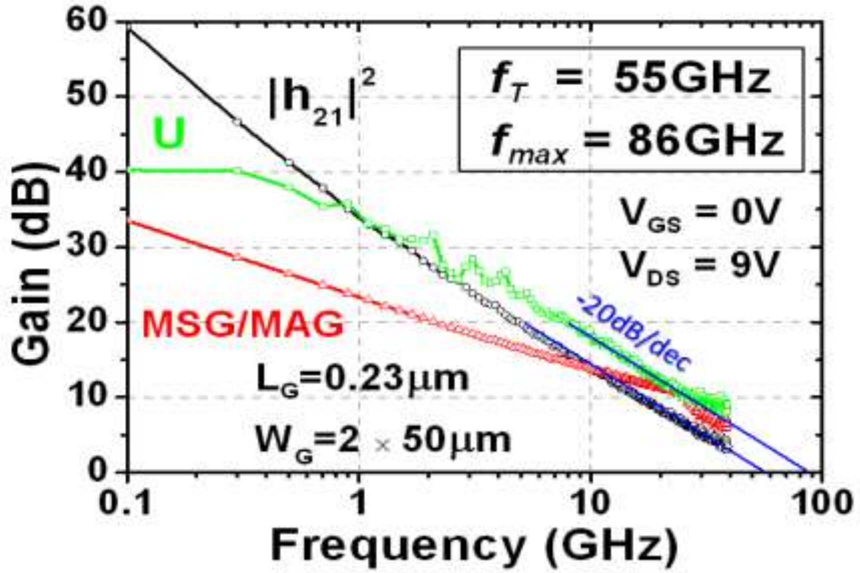
RF characterization & small signal modeling-2

- S-parameter & f_t/f_{max}

$$f_T = f(h_{21}=0) \qquad U(f) = \frac{\left|\frac{S_{21}}{S_{12}}-1\right|^2}{2\cdot k\cdot\left|\frac{S_{21}}{S_{12}}\right| - 2\cdot Re\left(\frac{S_{21}}{S_{12}}\right)}; \qquad f_{max} = f(U=0)$$



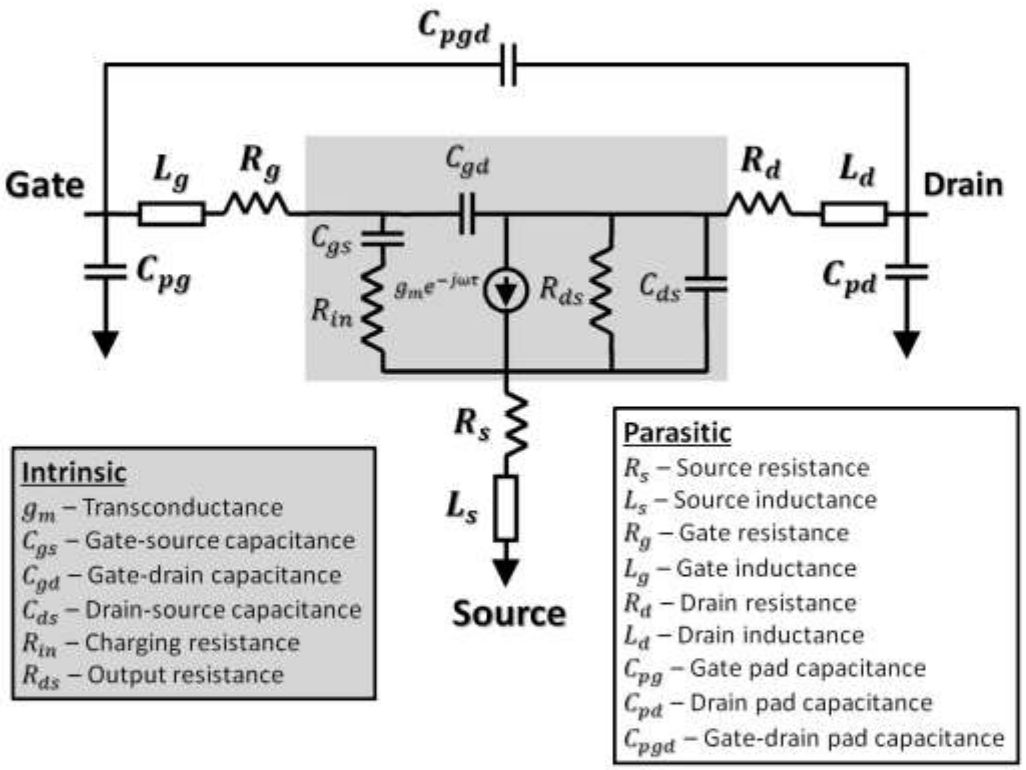
Smith chart shows the simulated and measured S-parameters

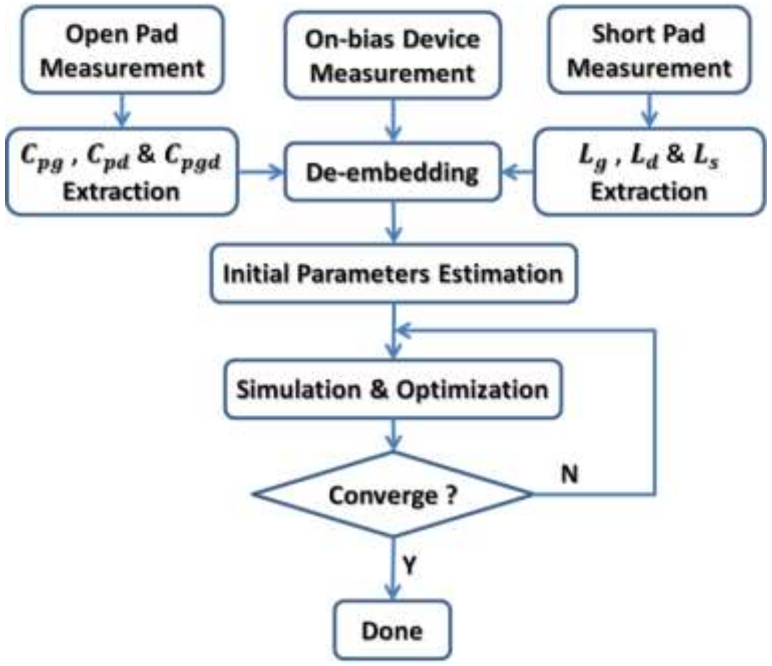


f_t & f_{max} can be obtained from the S-parameter measurements

RF characterization & small signal modeling-3

- The 15-element small signal equivalent circuit model



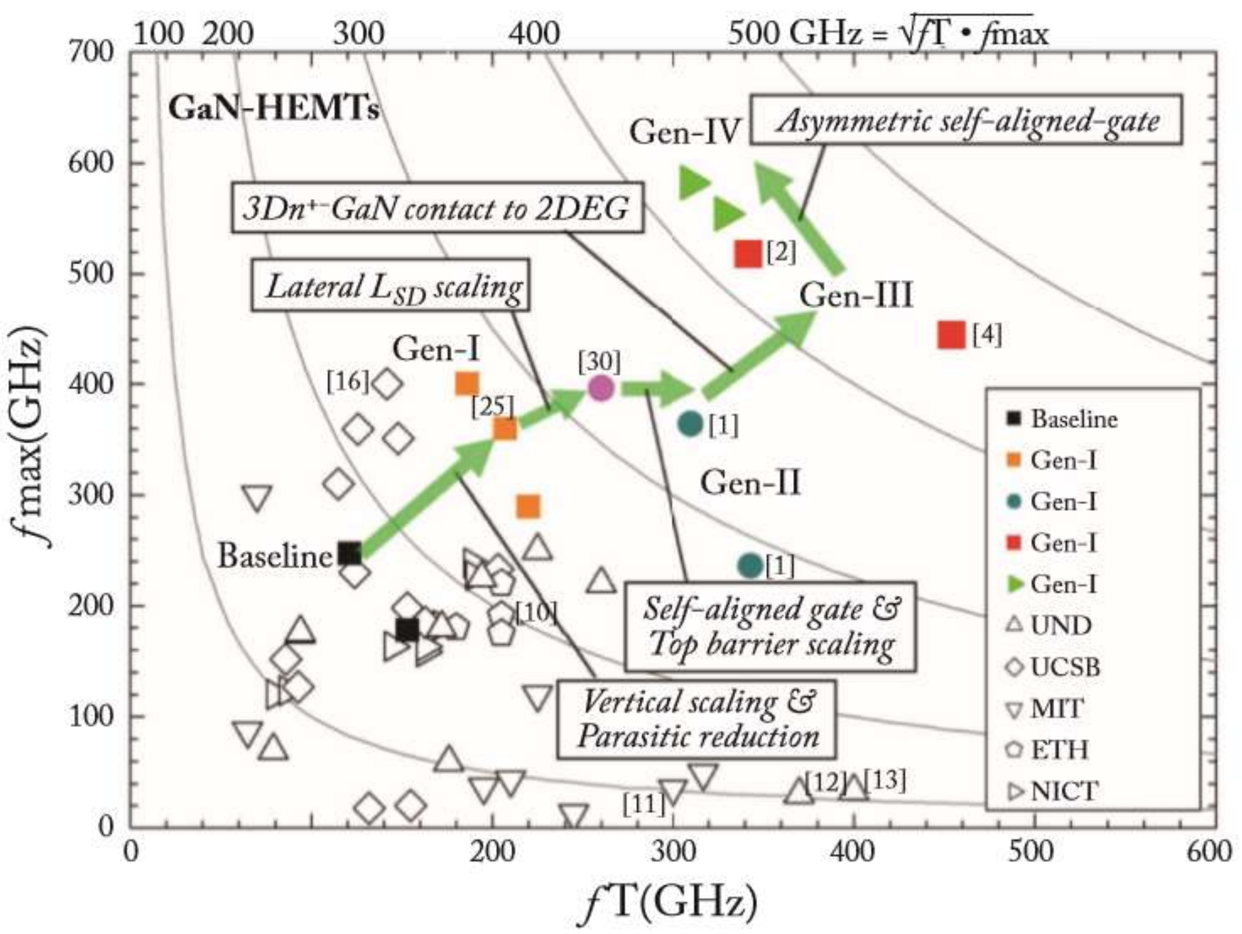


Approach of small signal equivalent circuit modeling

Advanced GaN device technologies

- Scaled gate length & T-gate → e-beam lithography
- Reduced contact resistance → heavily doped S/D regrowth
- Thin wide-band-gap barrier layers, such as lattice-matched InAlN or all-binary AlN, can simultaneously achieve a high channel sheet carrier density and reasonable gate control
- A T-shaped gate structure can simultaneously achieve a low gate resistance and a low parasitic gate capacitance

State-of-the-art GaN HEMT for RF

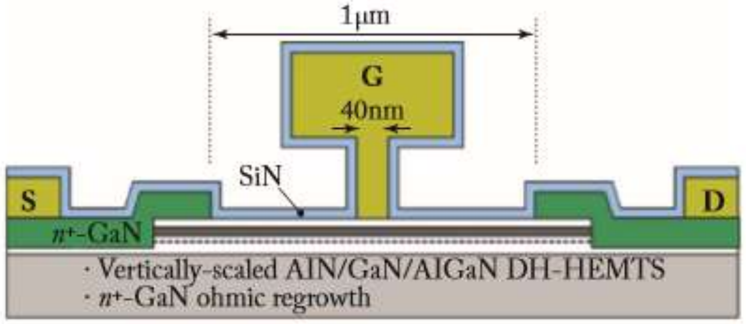


Summary of the RF performances for state-of-the-art GaN HEMTs in the literatures

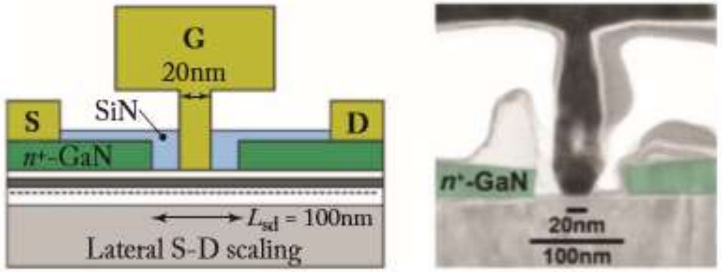
Both f_t & f_{max} exceeding 500 GHz

Ultra-scaled T-gate technology-1

- Cross-sections of four scaling generations of T-gate GaN HEMTs by HRL



(a)



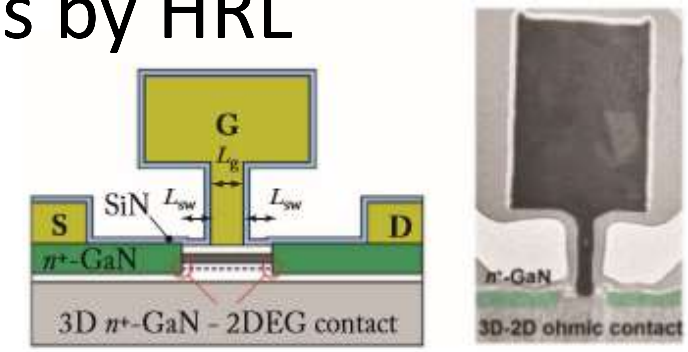
(b)

(a)Vertically-scaled AlN/GaN/AlGaN DH-HEMTS & n+-GaN ohmic regrowth

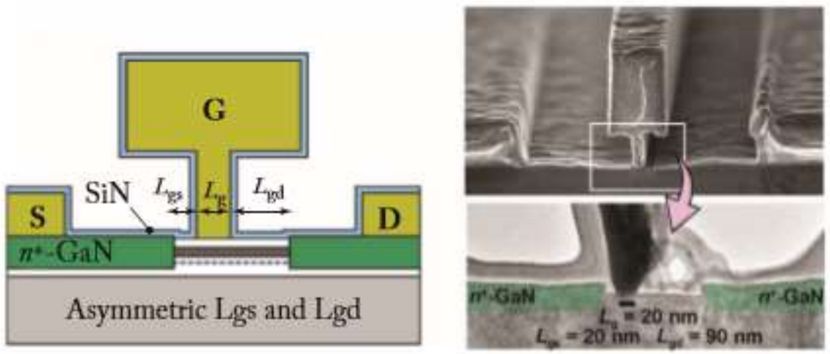
(b)Lateral S-D scaling

Ultra-scaled T-gate technology-2

- Cross-sections of four scaling generations of T-gate GaN HEMTs by HRL



(c)



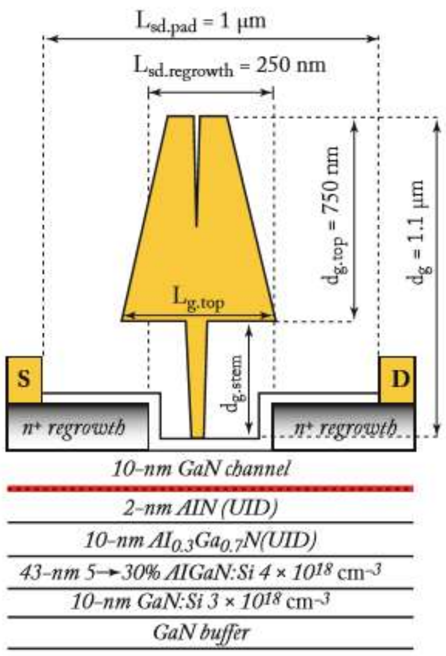
(d)

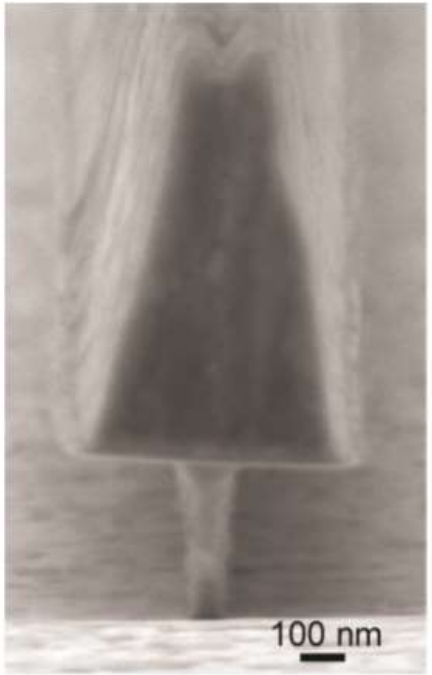
(c) 3D n+-GaN - 2DEG contact

(d) Asymmetric L_{gs} and L_{gd}

Ultra-scaled T-gate technology-3

- Cross-section of the MIS-HEMT with a high-aspect-ratio T-gate by UCSB (a high f_{max} of 351GHz for a 80 nm gate device)

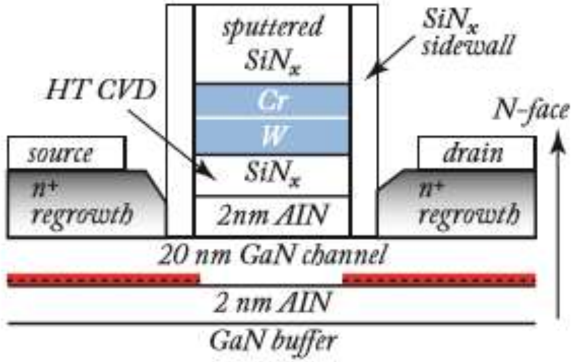




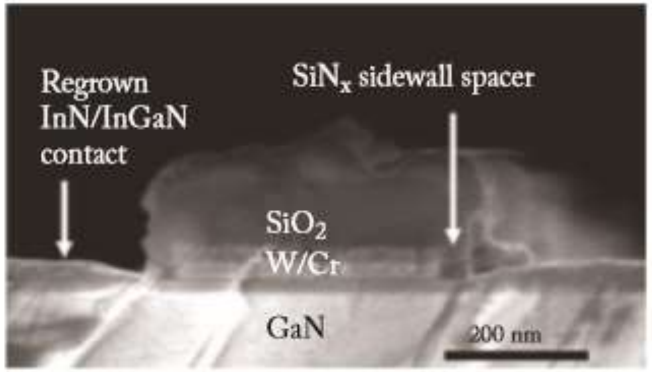
- An 80-nm-long, 1.1-um-tall Ti/Au T-gate with a 370-nm-tall stem formed by electron-beam evaporation and lift-off
- The purpose of the tall stem is to reduce parasitic capacitance under the top gate, whereas the purpose of the large top gate is to reduce the gate resistance

Self-aligned process-1

- Gate first



(a)



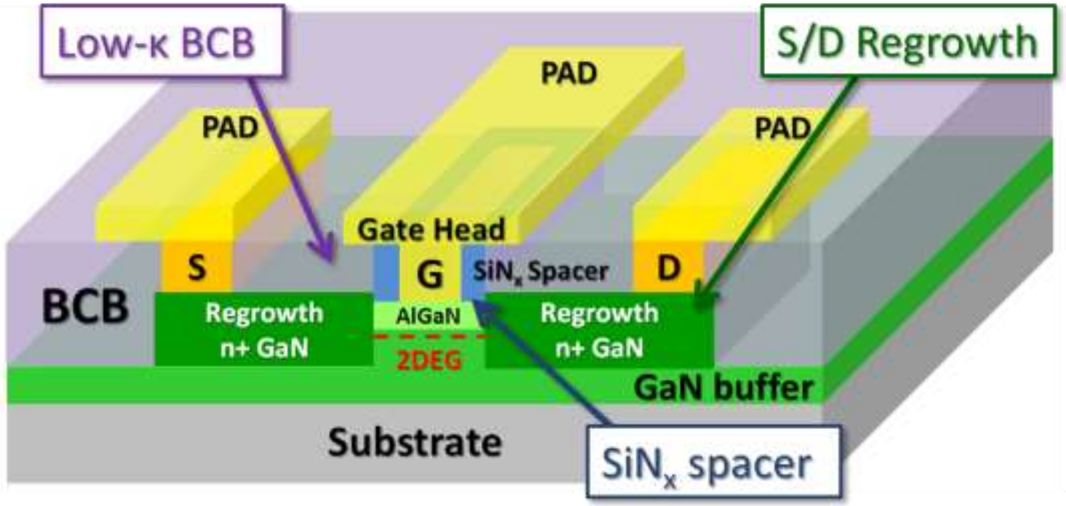
(b)

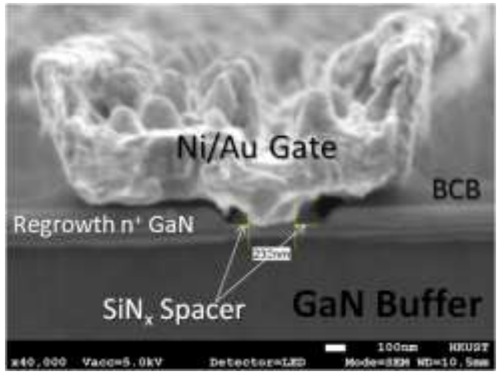
The process involves the deposition of a CVD SiNx gate dielectric followed by a W/Cr/SiO_2/Cr refractory gate stack, which is defined by e-beam lithography and patterned by dry etching. SiNx side wall spacers are formed by blanket deposition and an anisotropic dry etching process to isolate the gate metal from the regrown region. The spacer thickness controls the S/D access distances and the access resistances.

Disadvantage: high R_g, so low f_{max}

Self-aligned process-2

- Gate last



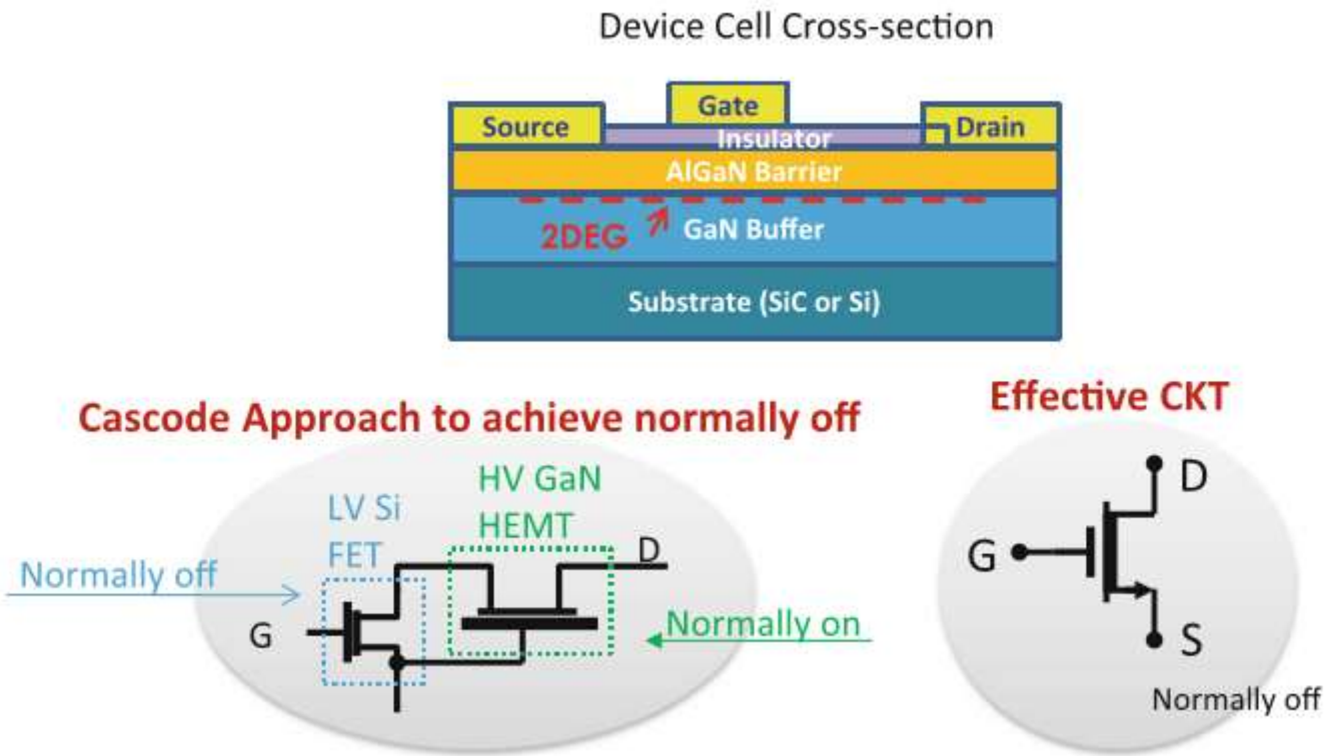


The gate-last self-aligned process was realized by a dummy gate, which was eventually removed after BCB planarization and replaced with a metal gate. Regrown ohmic contact and low-k BCB planarization technologies have enabled a reduction in the access resistance and in the parasitic capacitance, which minimizes RC-related delays.

GaN HEMT for power switch

- To minimize degradation of dynamic on-resistance and current due to electron trapping → passivation & material quality
- For fail-safe operation, power switches have to be normally off / Enhancement-mode (E-mode)
  - two-chip solution using a cascode connection of a depletion-mode device with a low voltage Si MOSFET (commercialized)
  - a single-chip solution by using a p-type gate to deplete the charge under the channel under zero bias in the GaN HEMT (commercialized)
  - other methods: gate recess or F-implantation (not commercialized)

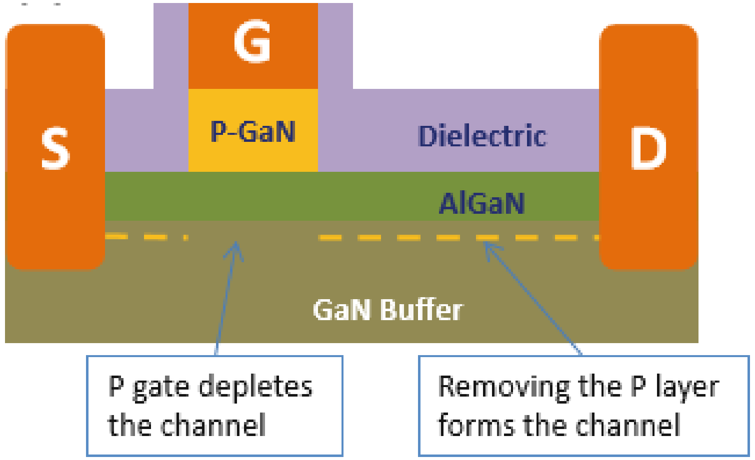
Cascode E-mode GaN power FET

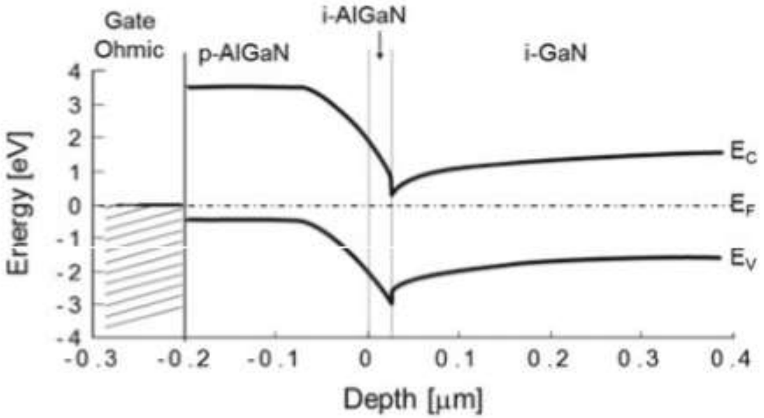


Cascode method for normally off operation. Disadvantage: two chips required

The Transphorm released 600 V rating cascode product

p-type gate E-mode GaN power FET





Band diagram at the gate region

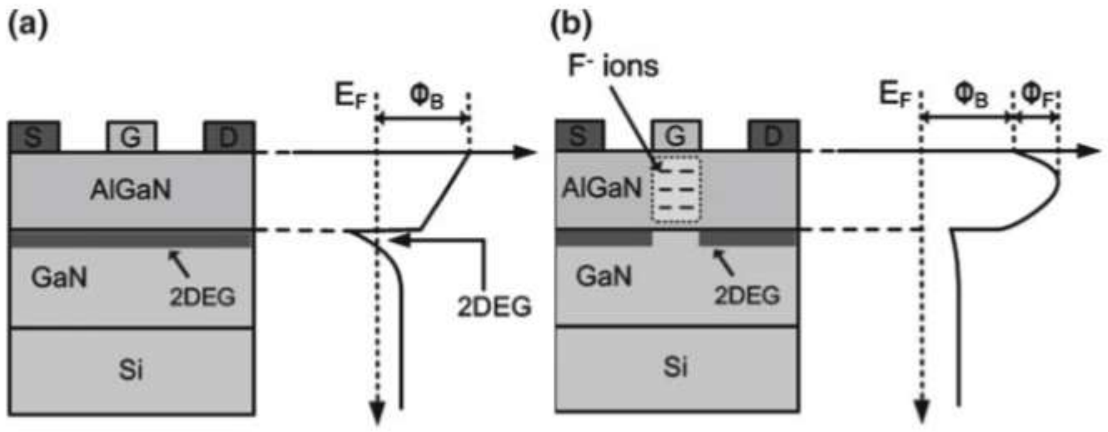
A single-chip solution by using a p-type gate to deplete the charge under the channel under zero bias in the GaN HEMT

Disadvantage: low threshold voltages

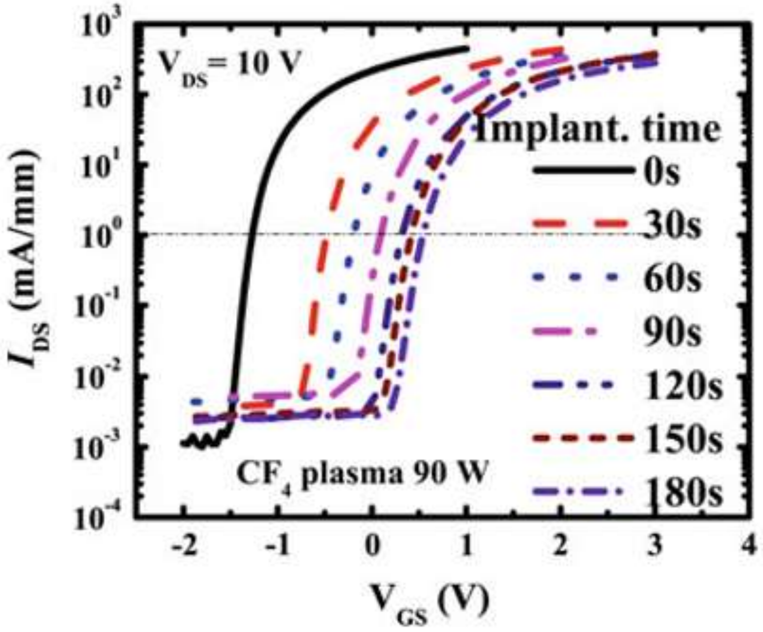
Products released by Panasonic, GaN systems and EPC

Other approach-1

- Fluorine-Implanted



Cross section and the conduction band profiles of standard normally on and normally off AlGaN/GaN HEMTs. The normally off HEMT features F− ions incorporated in the AlGaN barrier by plasma ion implantation

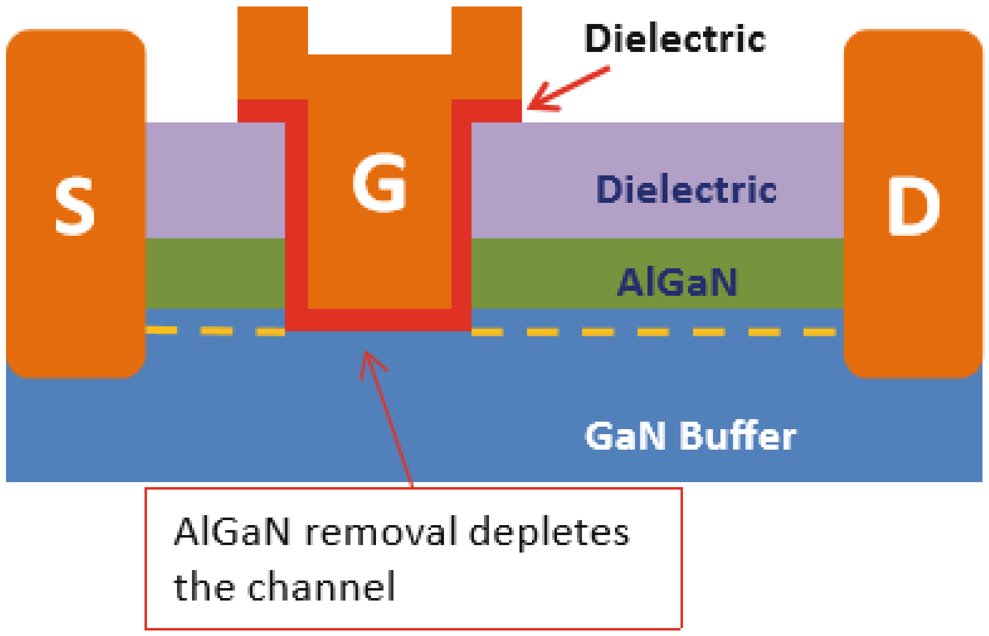


Transfer characteristics of AlGaN/GaN HEMTs with different fluorine plasma ion implantation times

Disadvantage： low V_{th}, stability & reliability issue

Other approach-2

- Gate recess



Challenges:

- etching control
- uniformity issues
- damage in the channel
- Reliable dielectric
- etc.

GaN applied to power electronics

- GaN device can switch at high frequency at high efficiency and low EMI allows the system to shrink their size and reduce the BoM cost of passive components and heat si
-

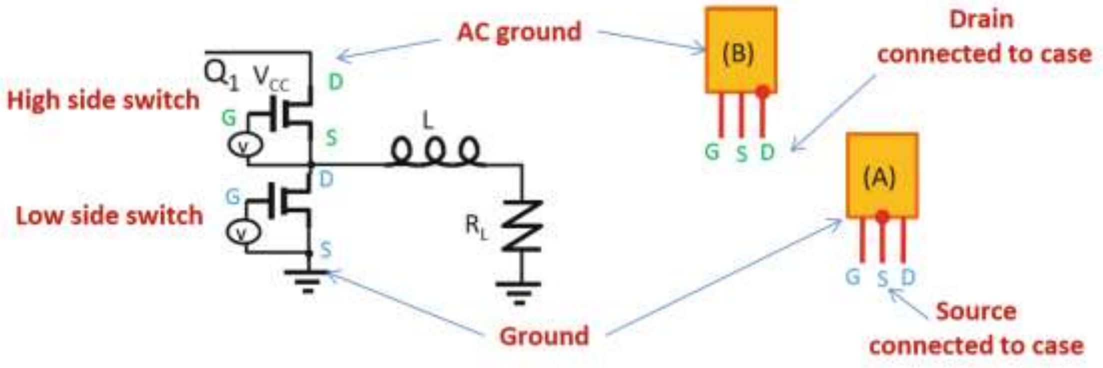


Fig. 4.22 Illustration of low-EMI packaging solution enabled by GaN

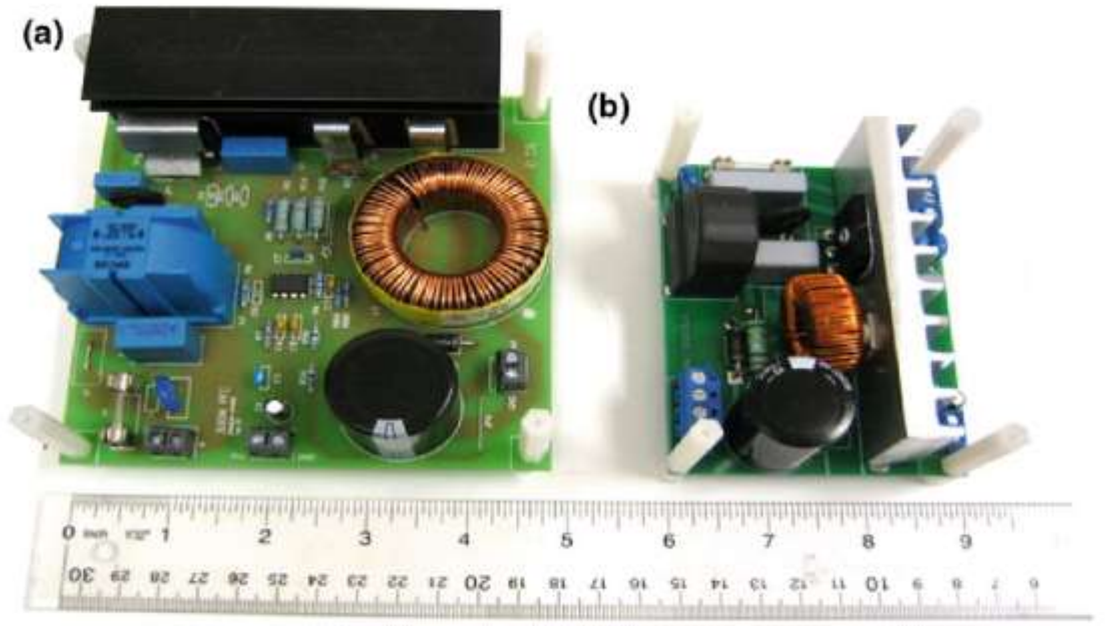


Fig. 4.23 Comparison of comparable PFC circuits: a silicon-based, requiring a two stage EMI filter and large inductors, and b GaN-based with smaller inductor size, one less EMI filter stage, enabled by the lower-loss transistors and diodes

What you can take away from Chapter 6

- Concept of GaN HEMTs and 2DEG
- Device structure & band diagram
- Fabrication process
- Key issues such as breakdown, reliability, leakage current, frequency. Etc…